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# Week 7:

## Decisions under uncertainty part 2 + Risk

**Instructor:** Brielle K Thompson

**Course:** NAT_R 8001 Decision Analysis for Research and Management of Natural Resources

# Begin with a few final presentations

# Review of last week

- Background on types of uncertainties
  - Linguistic vs Epistemic uncertainties
- Learned about how sensitivity analysis, value of information, and adaptive management can be used in decisions where uncertainty may impede management
  - **Sensitivity analysis**
    - Perturbations to understand whether uncertainty is important to resolve
  - **Value of information**
    - Analysis to quantify the improvement of an outcome if uncertainty is resolved
  - **Adaptive management**
    - Formal process of “learning by doing” for repeated decisions

Today:
How do we make decisions under uncertainty & risk?

# Working with uncertainty

Three Options:

1. Make decisions anyway
   Risk tools: e.g., Decision Trees, Utility Theory
2. Conduct research to reduce uncertainty
   Value of Information
3. Learn while managing
   Adaptive management

**Risk = topic for this week!**

**last week**

First, I'll cover a type of quantitative approach for adaptive management called Management Strategy Evaluation

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# Learn while managing: adaptive management

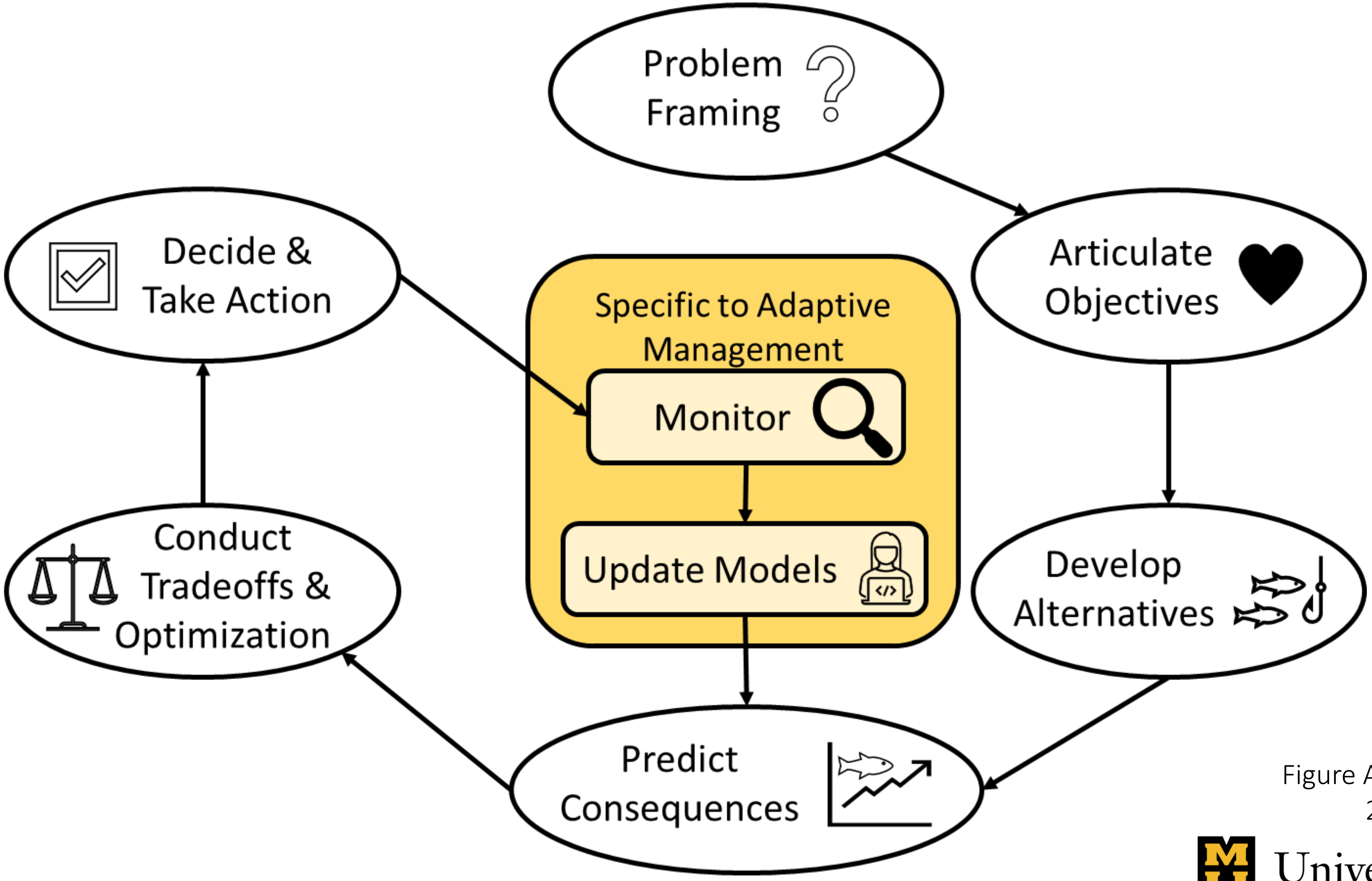


Figure Adapted from Converse et al. 2013 and Runge 2011

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**Degrees of monitoring:**
- Monitor single management action
- Monitor multiple management ‘experiments’

**Degrees of updating models:**
- Simple as ‘recording new data’
- Complex as Bayesian updating

**Degrees of predicting consequences:**
- Expert elicitation
- Prediction modeling
- Experiments
- Management strategy evaluation

**Degrees of tradeoff & optimization:**
- Negotiation
- Multi-criteria decision analysis
- Value of information
- Dynamic programming methods*

**beyond scope of class*

# Specific methods used:

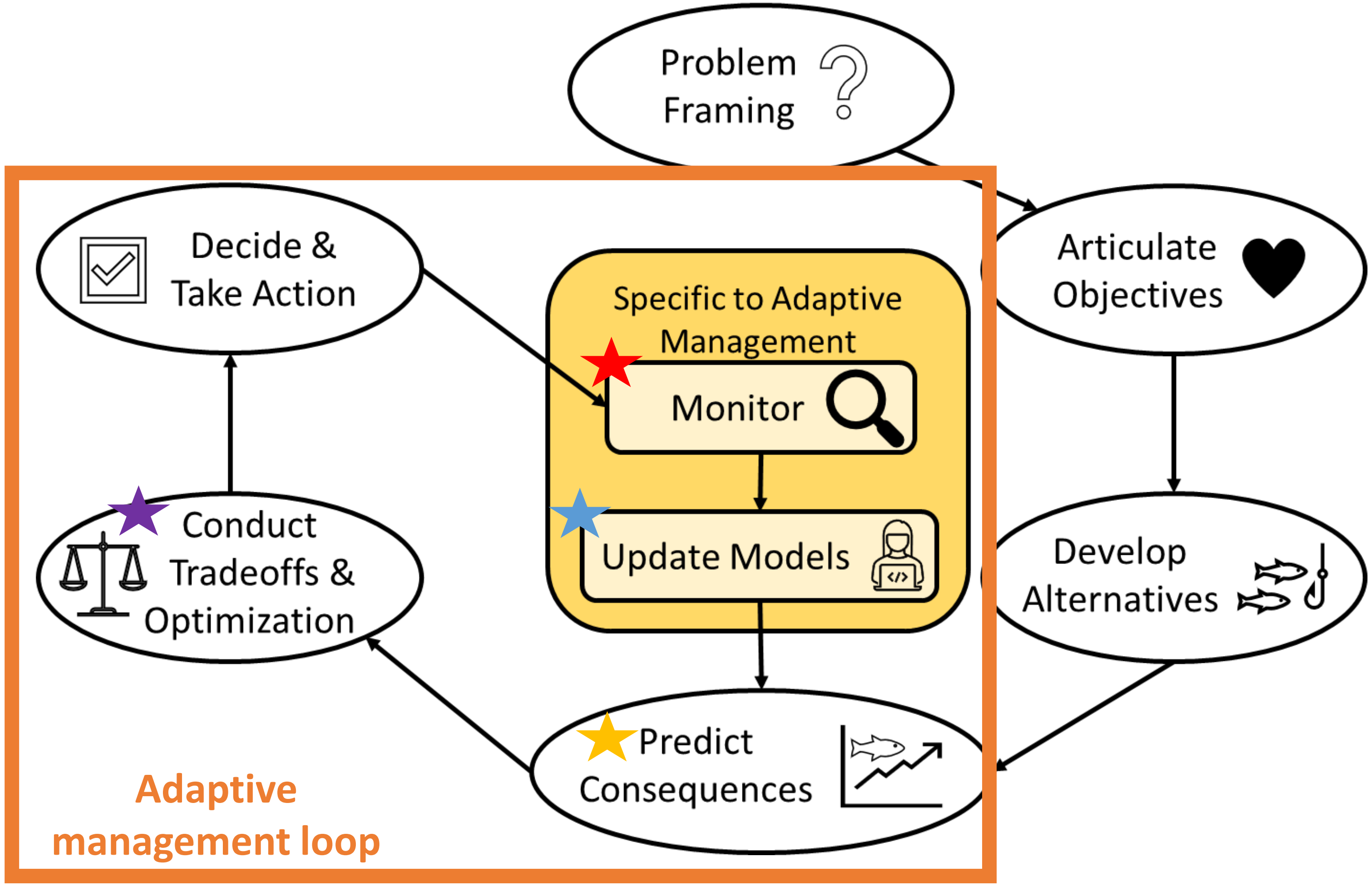


Figure Adapted from Converse et al. 2013 and Runge 2011

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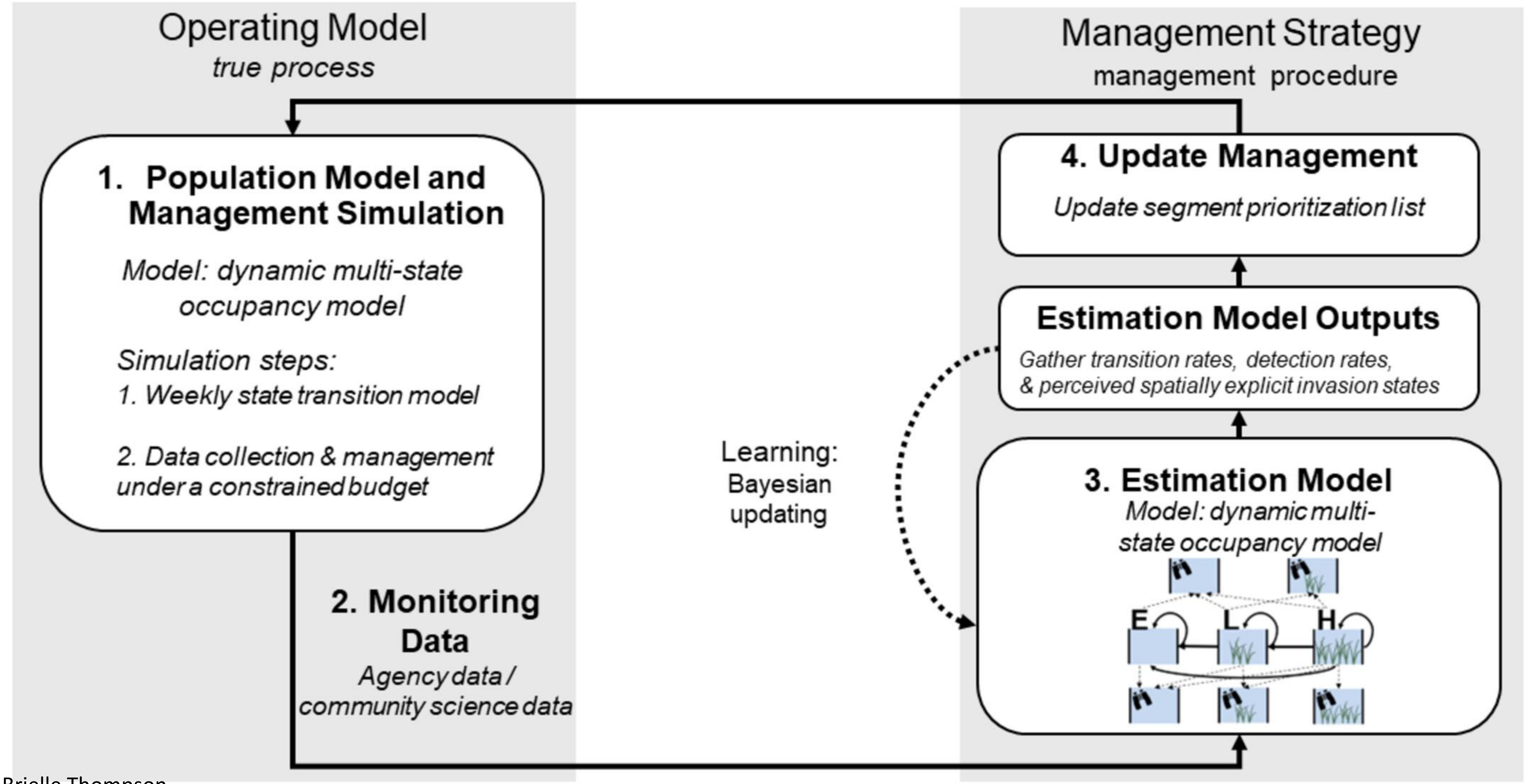
# Another useful adaptive management framework:

## Management strategy evaluation

Thompson et al. 2025

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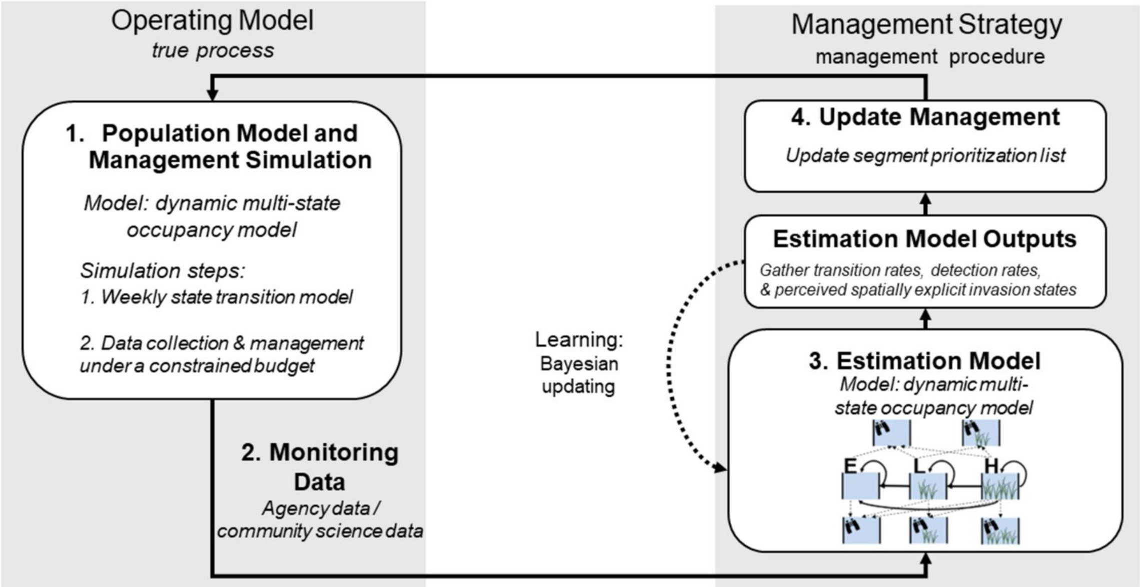
# Management Strategy Evaluation

- Approach for identifying optimal adaptive management decisions
- Twophases:
  - **Operating model**
    - Underlying ecological process
    - Results in monitoring data
  - **Management strategy**
    - Uses monitoring data to inform estimation model
    - Estimation model then informs management decision
- Use:
  - Fisheries, invasive species, large predator management

Thompson et al. 2025




**Pros:** can incorporate complex ecological processes in estimation/simulation models

**Cons:** Does not inform mathematically optimal decisions (compared to dynamic programming processes)

# Working with uncertainty

Three Options:

1. Make decisions anyway

   Risk tools: e.g., Decision Trees, Utility Theory

2. Conduct research to reduce uncertainty

   Value of Information

3. Learn while managing

   Adaptive management

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# Assessing risk

- Making a decision under uncertainty involves the decision maker's risk perception and attitude towards that risk

- Perception: a subjective judgment of risk

- Attitude: an individual's tendency to avoid or seek risks

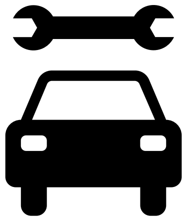
# Risk attitude

- The decision-maker’s risk attitude reflects their values

- Often reflect a public value, i.e., policy
  - Legal mandates (e.g., ESA)
  - Western science (5% p-values)

# Risk attitude examples

## Risk-averse



- Buying insurance
- You would pay a sure amount to avoid a gamble with an average loss that is smaller than the sure payment. Why do it?

## Risk-seeking


- Buying a lottery ticket
- You would pay a sure amount to enter a gamble with a smaller average return than the sure payment. Why do it?

## Risk-neutral

- Choosing a standard investment with average returns (e.g., S&P 500)
- You would put an investment in something that on average will give you good returns

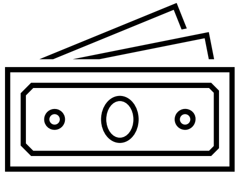
**risk averse investment = bonds/high-yield savings*

**risk seeking investment = individual stocks, crypto*

# Risk attitude: examples

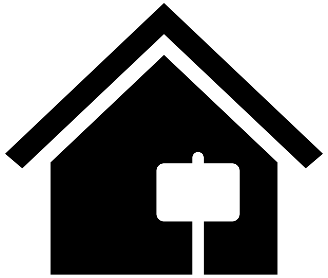
## Risk-averse

- Accepting a salary instead of commission-based pay



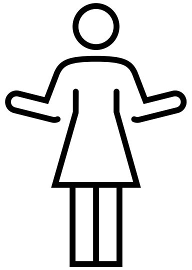
## Risk-seeking

- Chooses commission-based pay hoping for a big payout despite risk



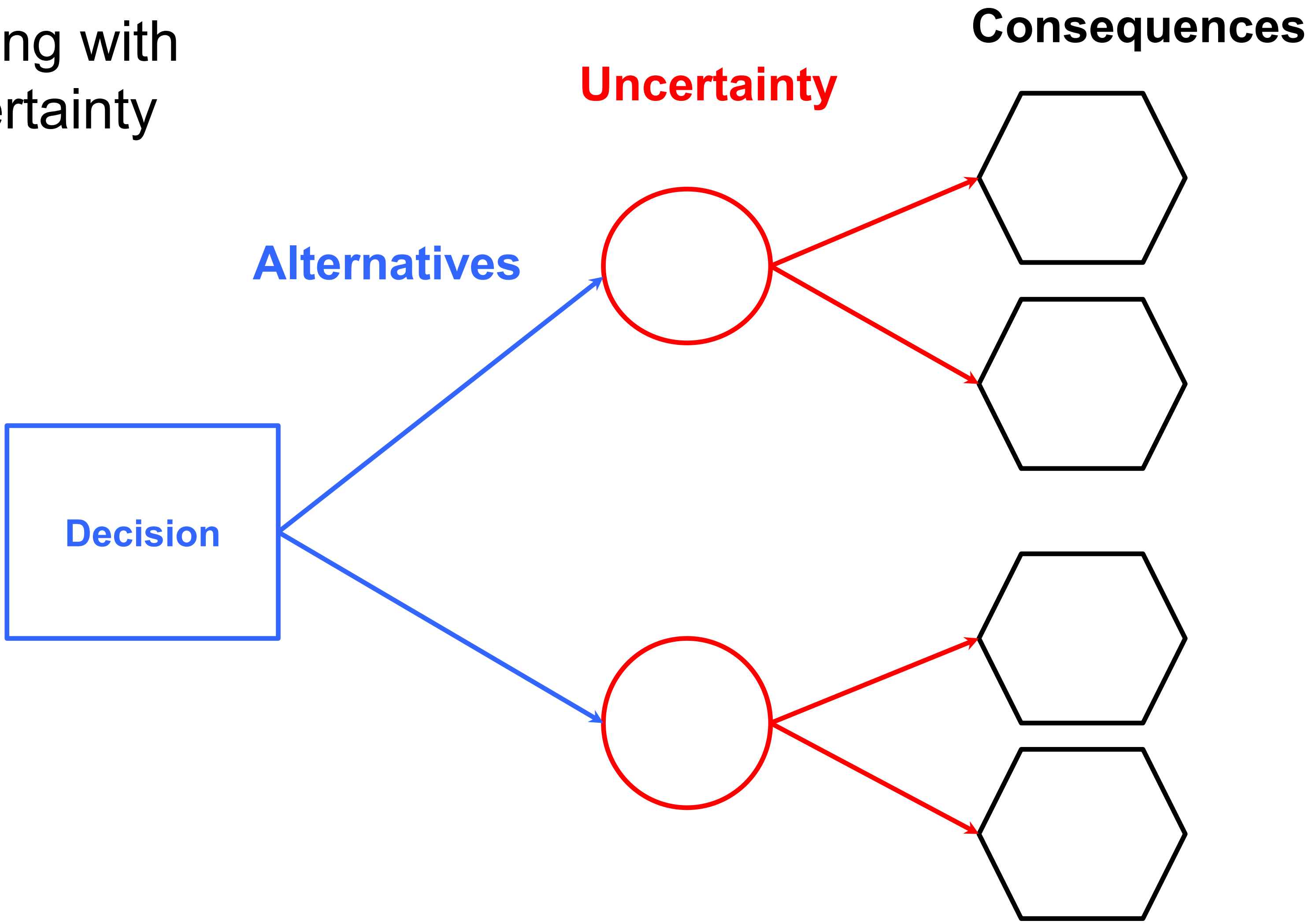
## Risk-neutral:

- Indifferent between salary or commission as long as average payout is the same



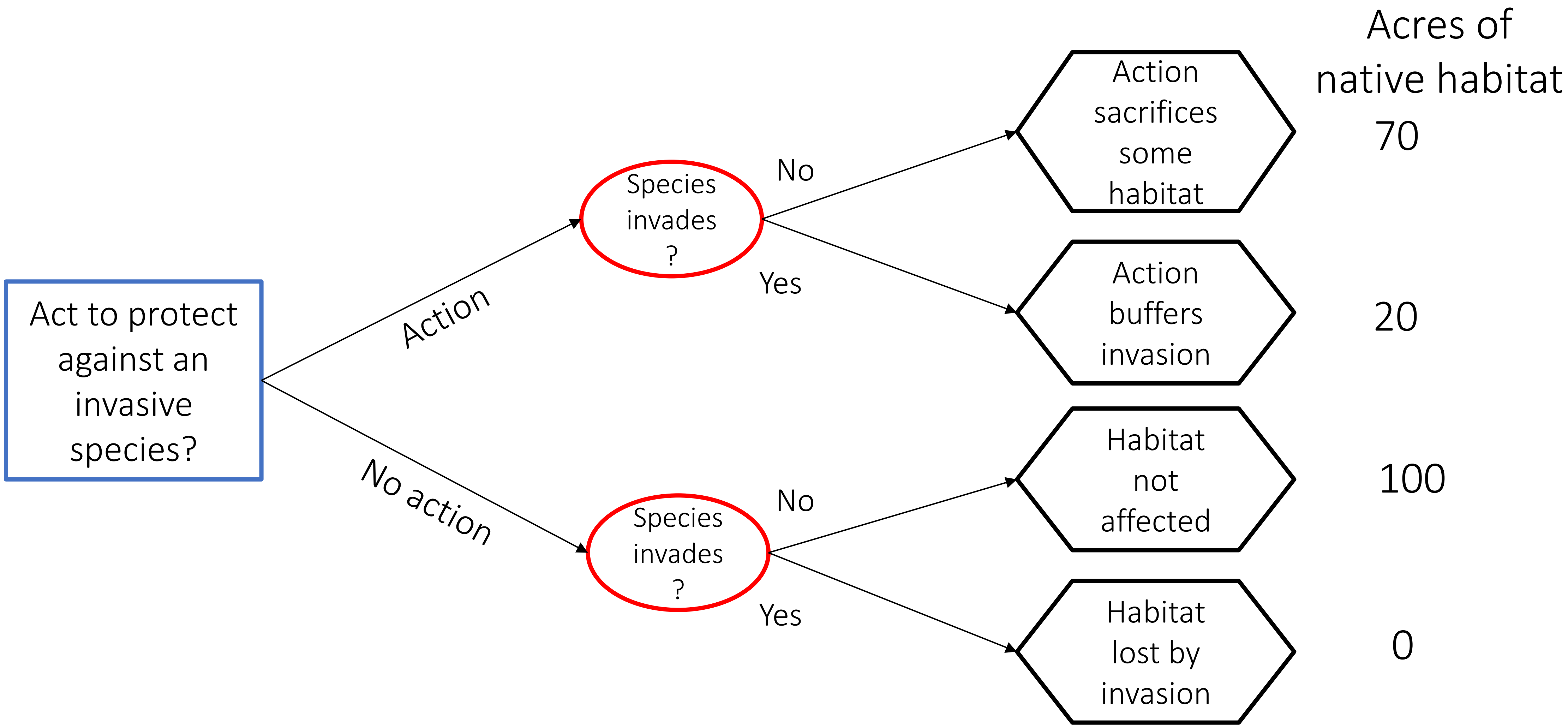
# Back to decision trees:

A useful tool for dealing with uncertainty



# Decision tree example:

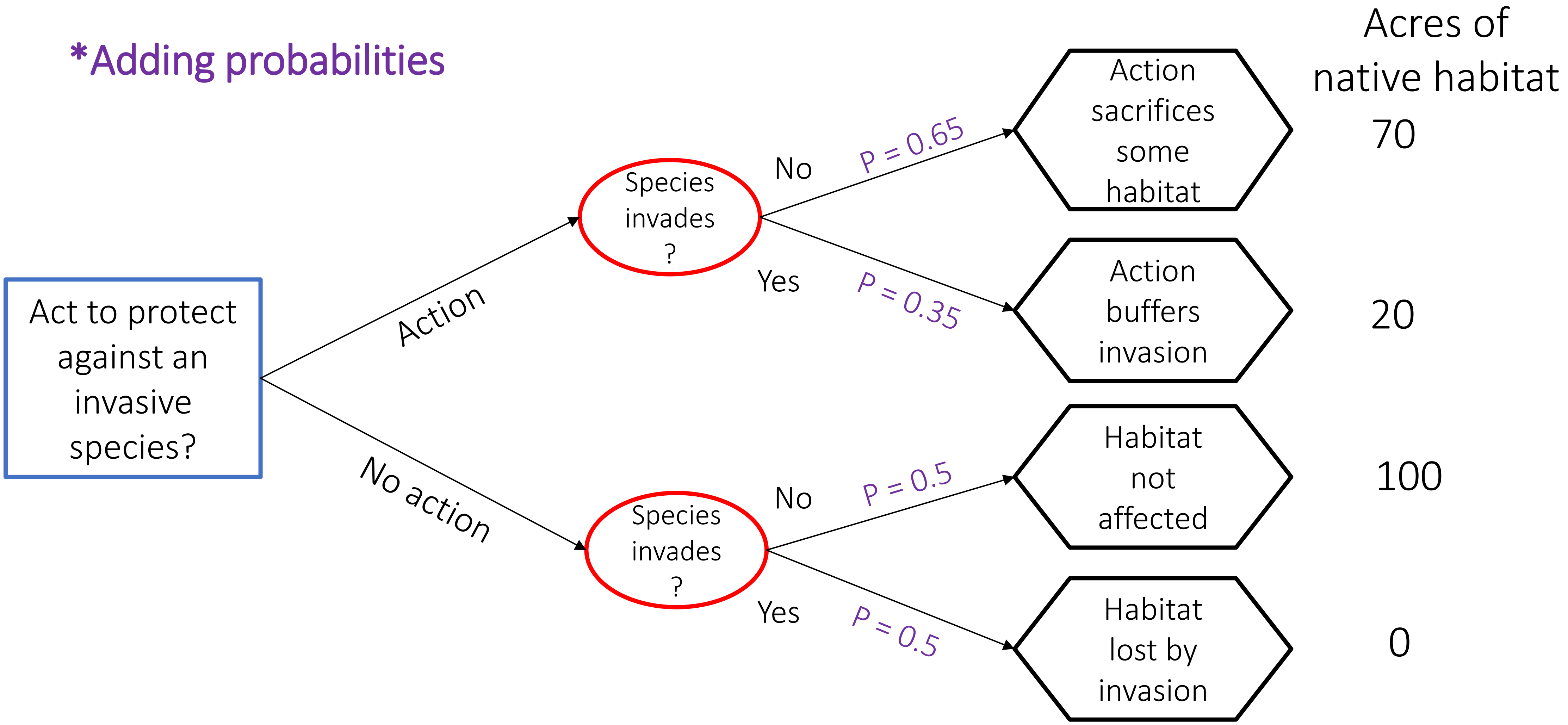
**Decision = Whether or not to create a no-habitat buffer around the refuge**



# Decision tree example:

Decision = Whether or not to create a no-habitat buffer around the refuge

*Adding probabilities

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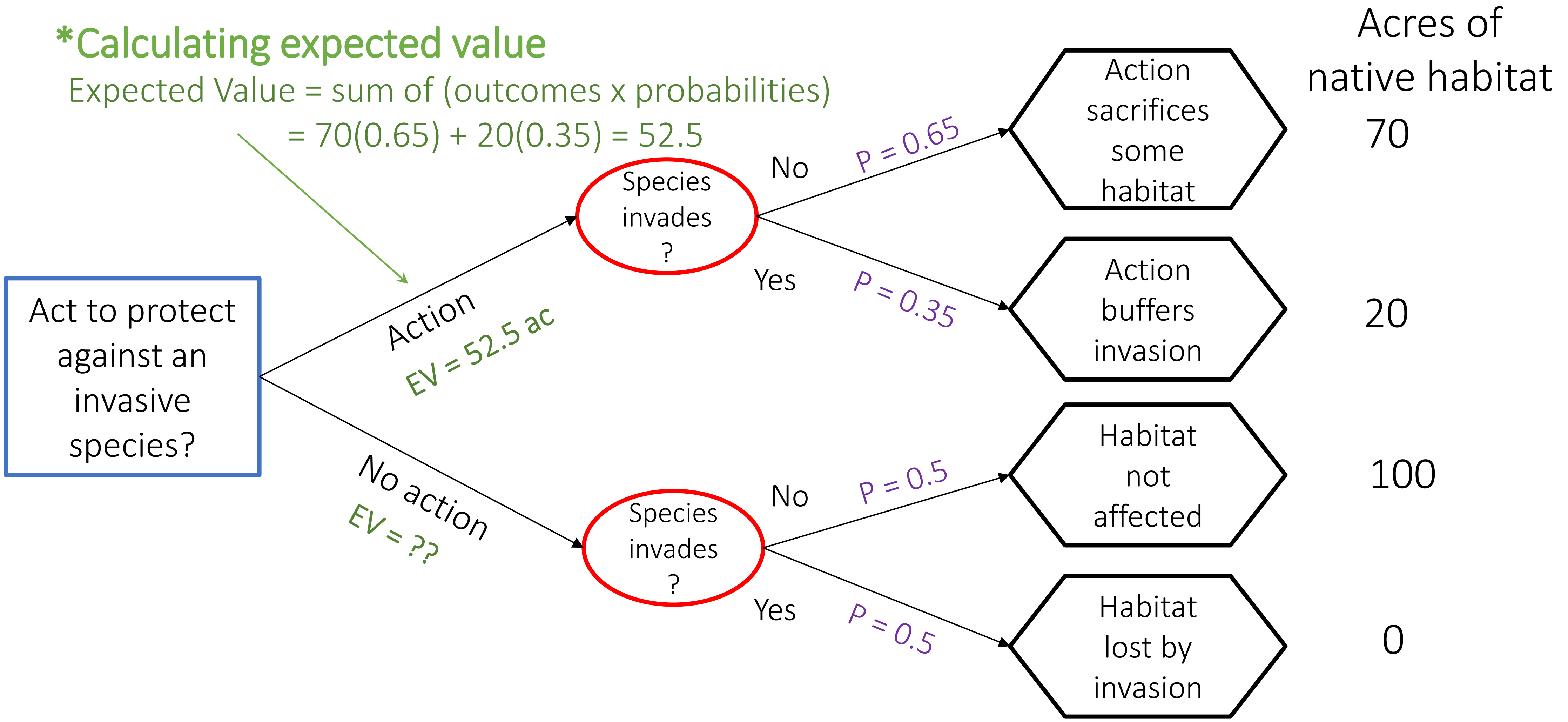
# Decision tree example:

## Decision = Whether or not to create a no-habitat buffer around the refuge

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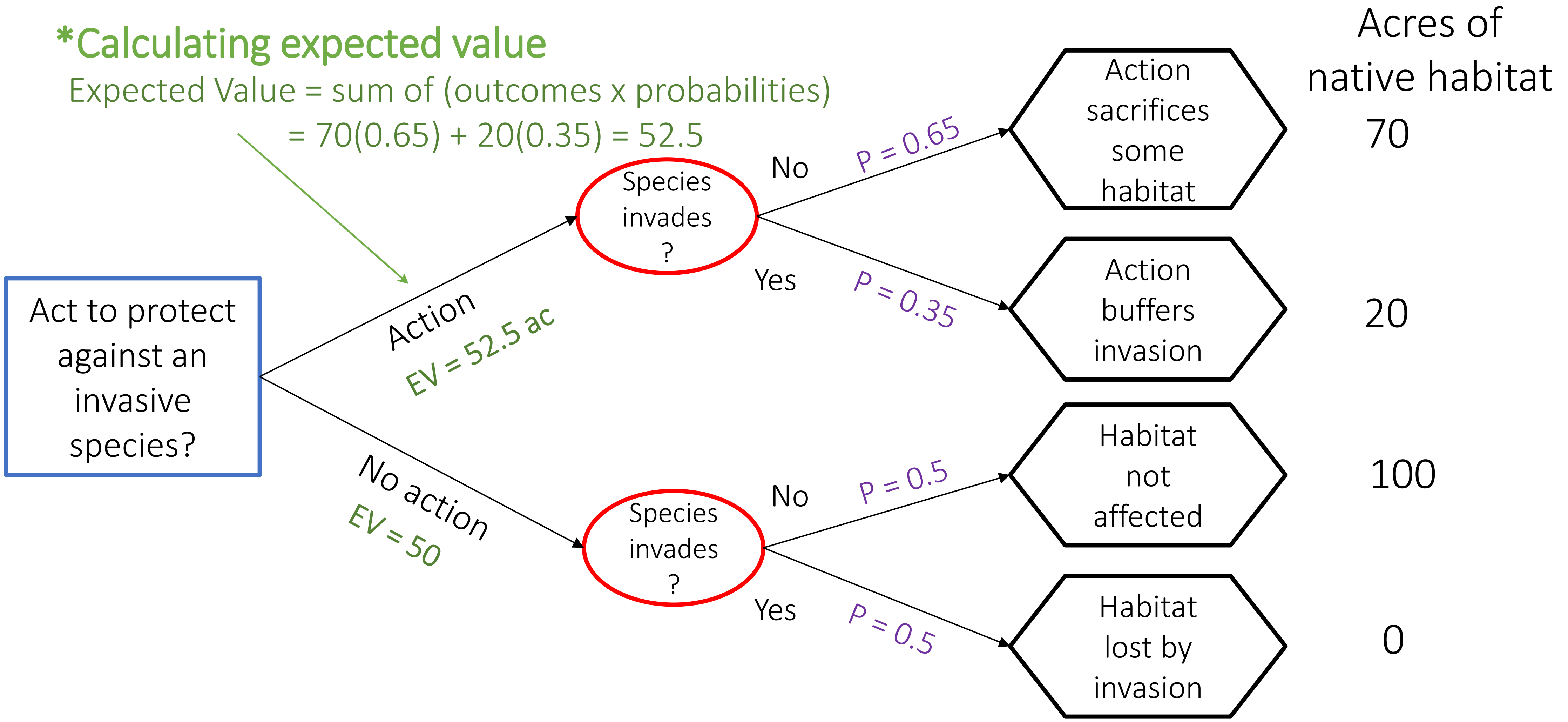
# Decision tree example:

Decision = Whether or not to create a no-habitat buffer around the refuge



# Activity: Your risk attitude

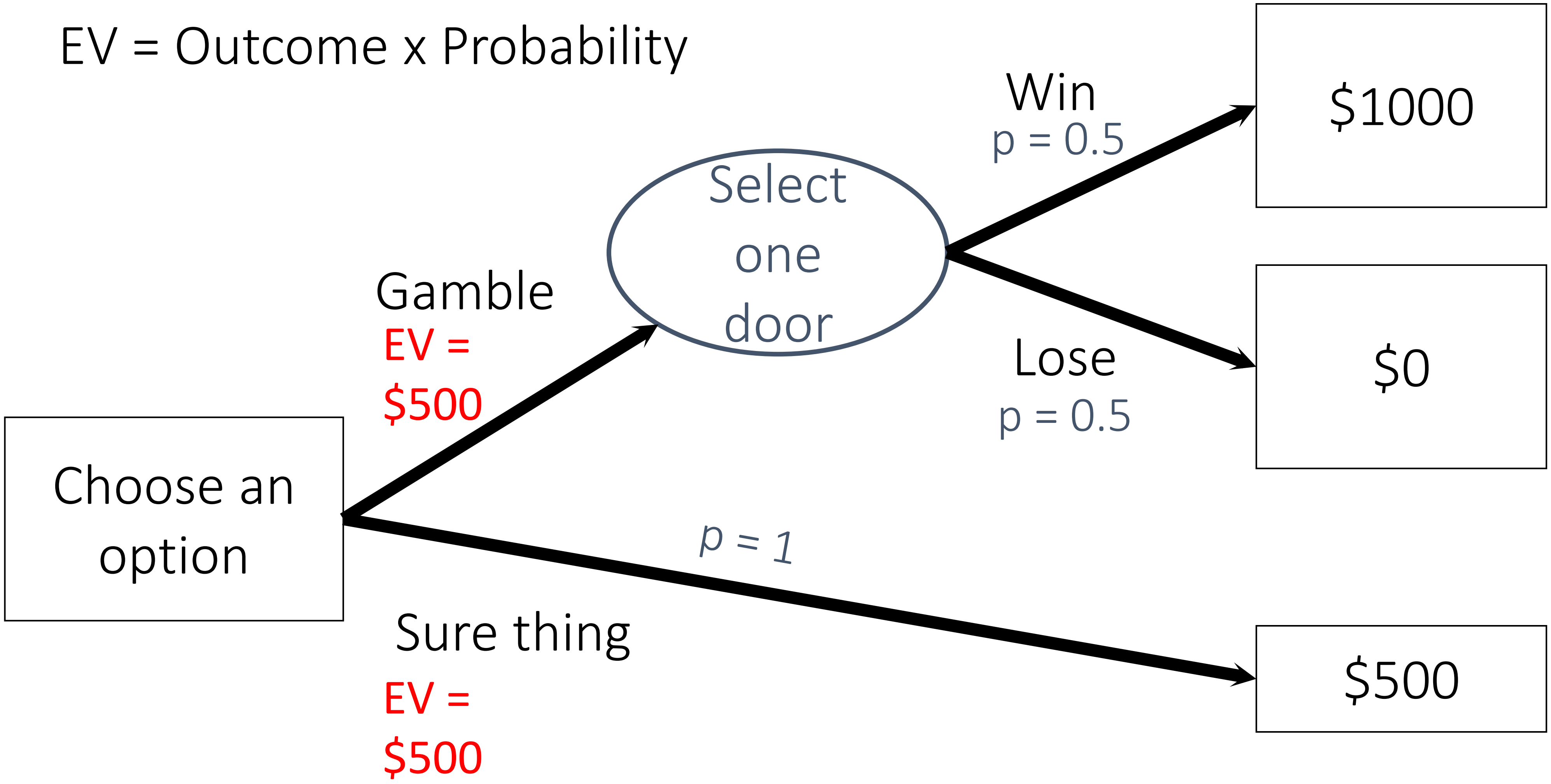
- Consider the following game show with two options:
  1. **Take the gamble:** Select one of two doors and claim a prize. Behind one door is $1000 and behind the other is $0.
  2. **Take the sure thing:** Take $500 instead.

- Which option would you choose?

- Draw a tree to represent your options.

# Activity: Your risk attitude

EV = Outcome x Probability



# Approaches to risk:

- Expected value (EV)
  - Method = outcomes x probabilities
  - Solution = maximize EV
- Expected utility (EU)
  - Method = Translate outcome onto a scale that incorporates risk attitude
  - Solution = Maximize EU
- Other approaches
  - Mini-max: Minimize risk of large losses
  - Robustness: Focus on attaining a minimum performance requirement over greatest range of uncertainty



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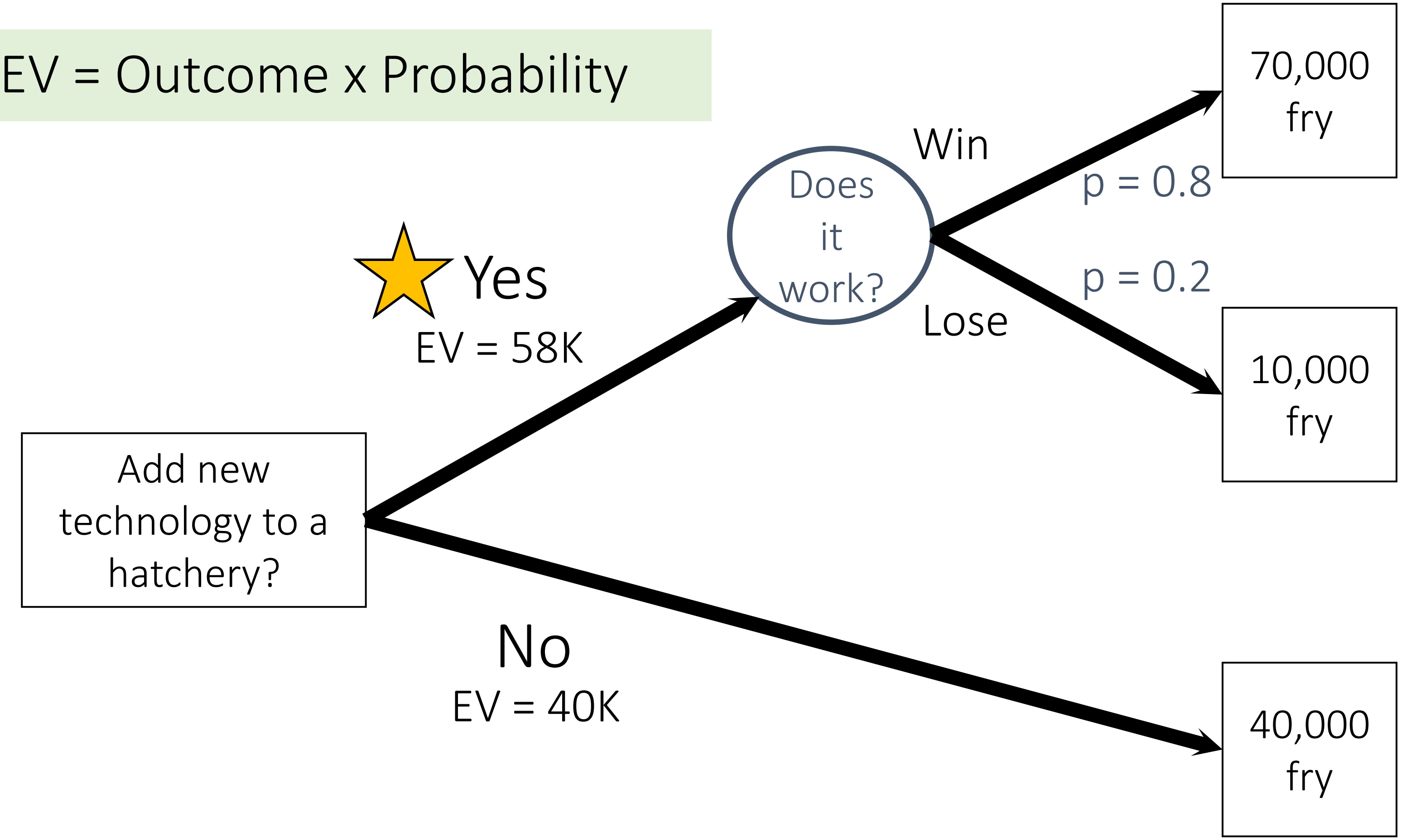
# Approaches to risk example:

- Should you adopt new technology to increase production of fry at a hatchery?

EV = Outcome x Probability



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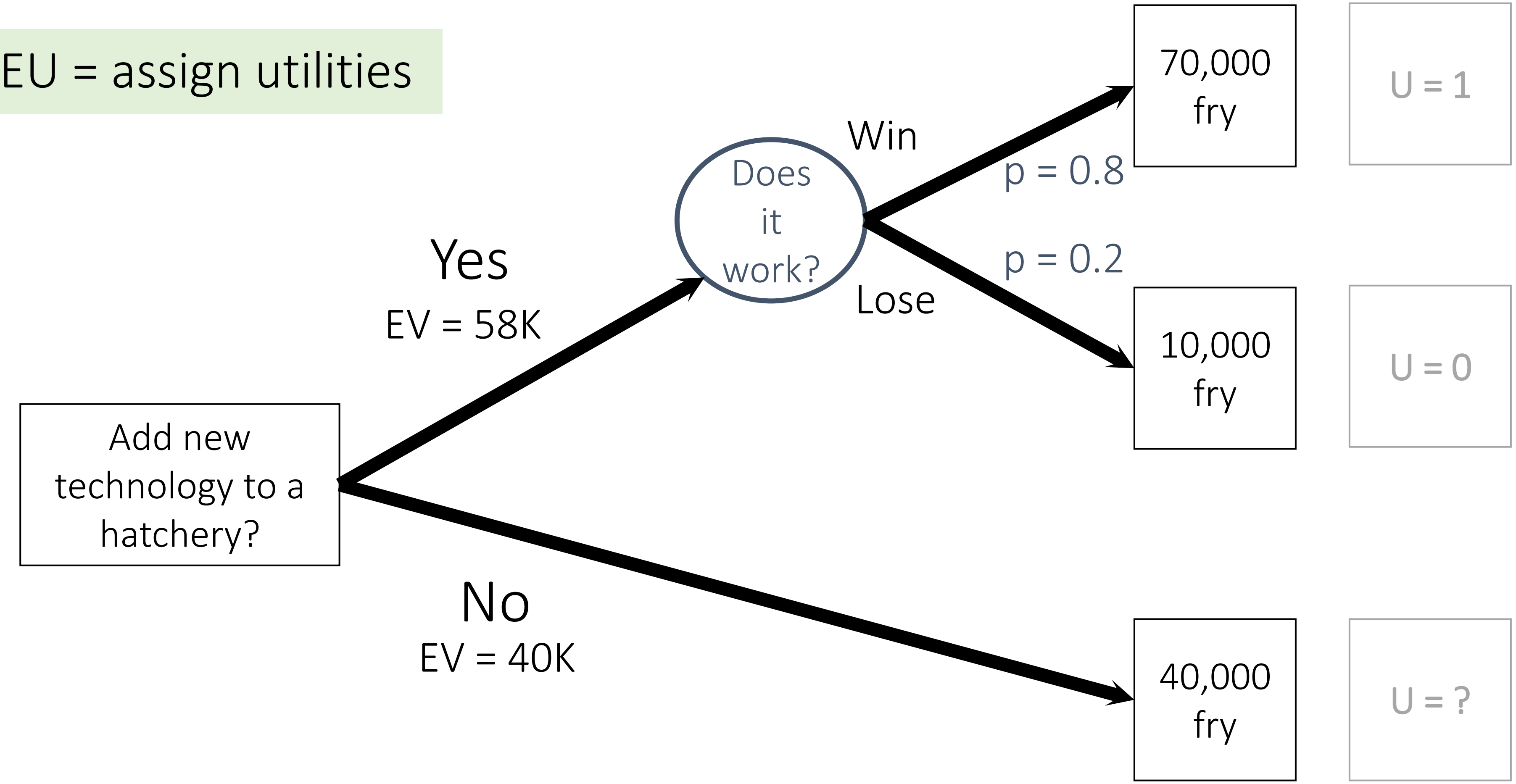
# Expected utility:

- Useful for non-neutral risk attitudes
  - Convert outcome to a utility score to represent attitude toward risk
  - e.g., winning $1000 has U=1.0, $500 in pocket has U=0.6
- Can be applied to multiple objectives
- There are a variety of ways to determine utility

# Approaches to risk example:

- Should you adopt new technology to increase production of fry at a hatchery?

EU = assign utilities




*Assign highest outcome utility of 1

*Assign lowest outcome utility of 0

?????
40K is halfway between 10K and 70K

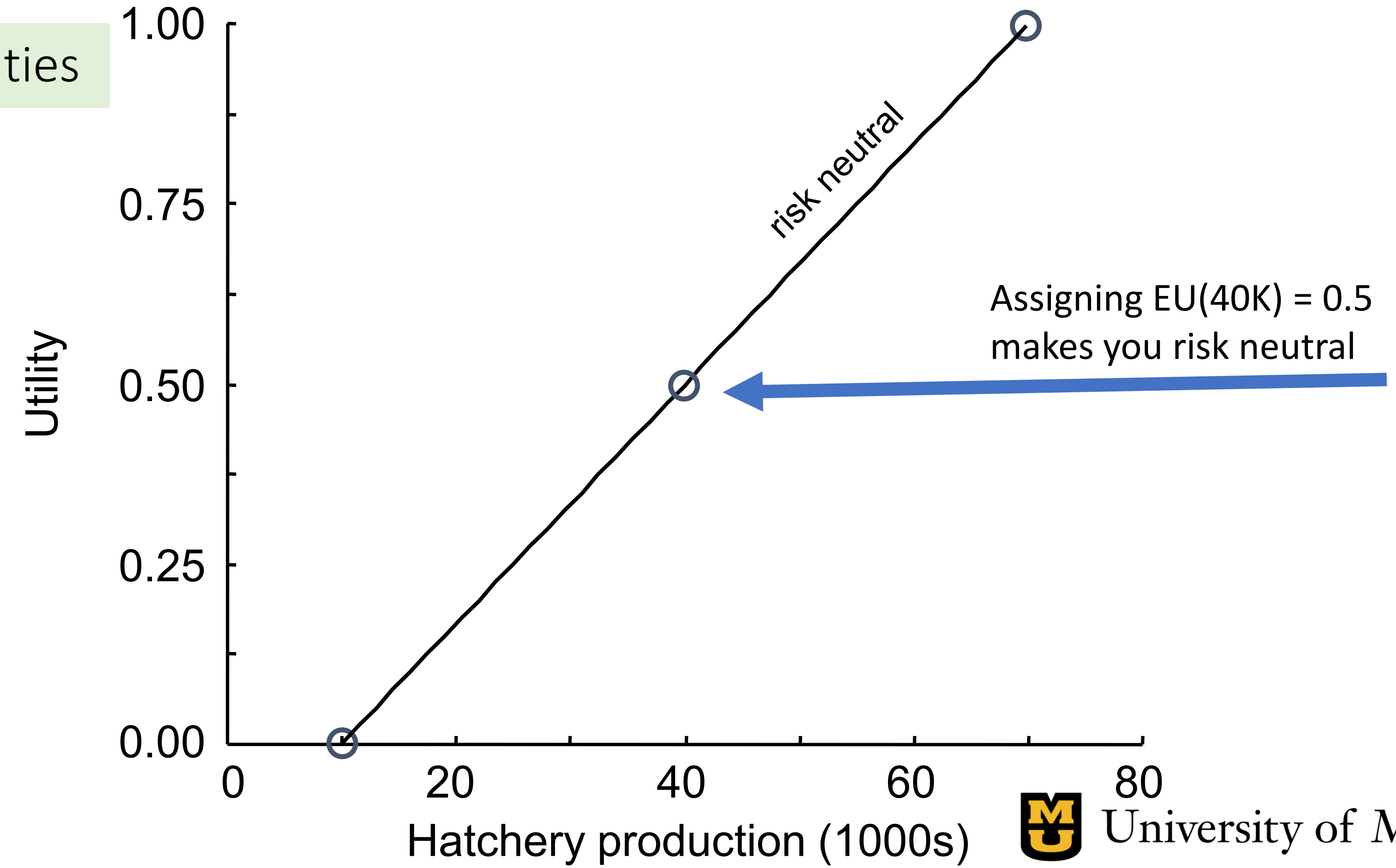
# Approaches to risk example:

- Should you adopt new technology to increase production of fry at a hatchery?

EU = assign utilities

<u>Utility of 40 K?????</u>

**40K is halfway between 10K and 70K**



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# Approaches to risk example:

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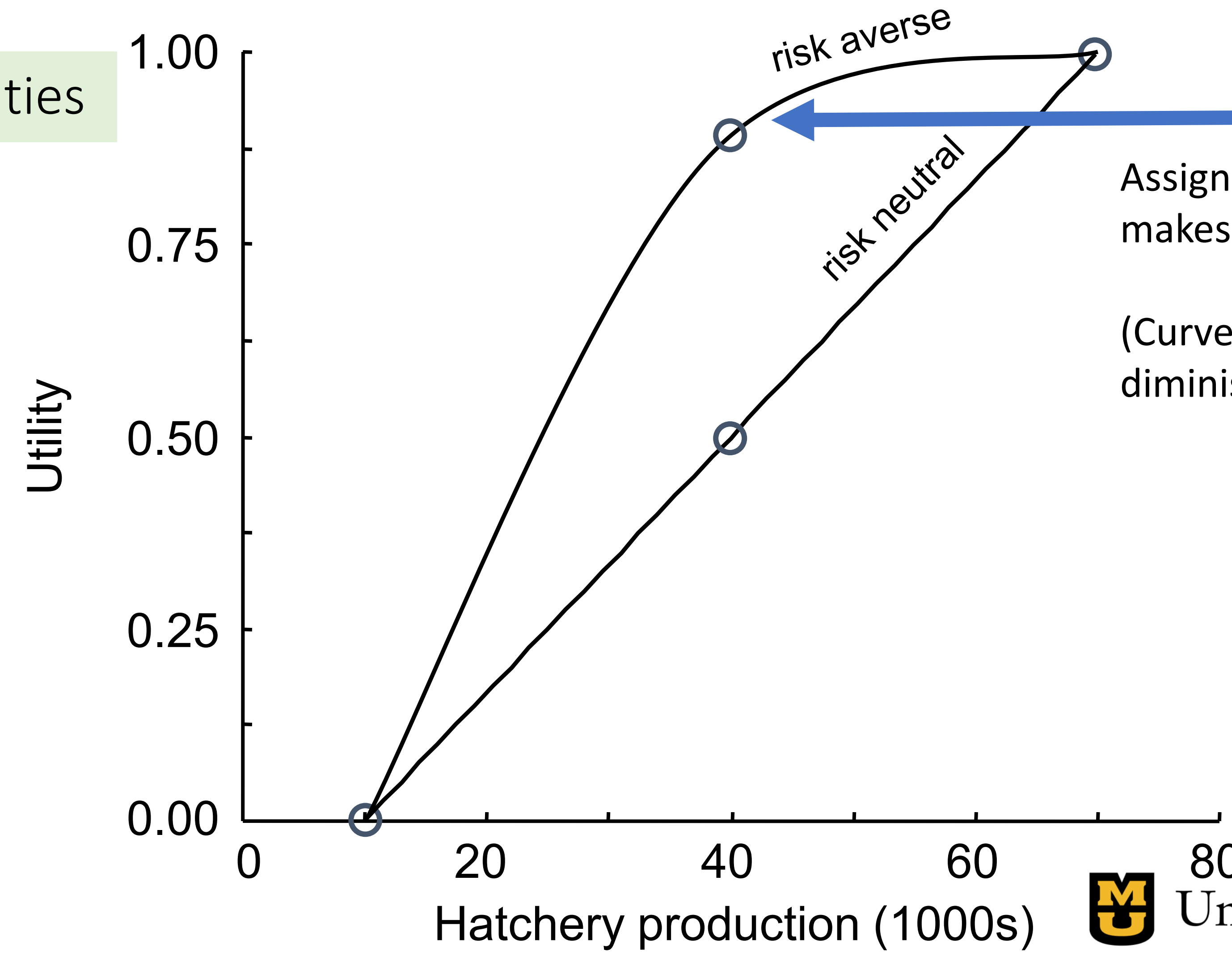


Assigning EU(40K) > 0.5 makes you risk averse

(Curve is concave showing diminishing marginal utility)



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# Aside: diminishing utility


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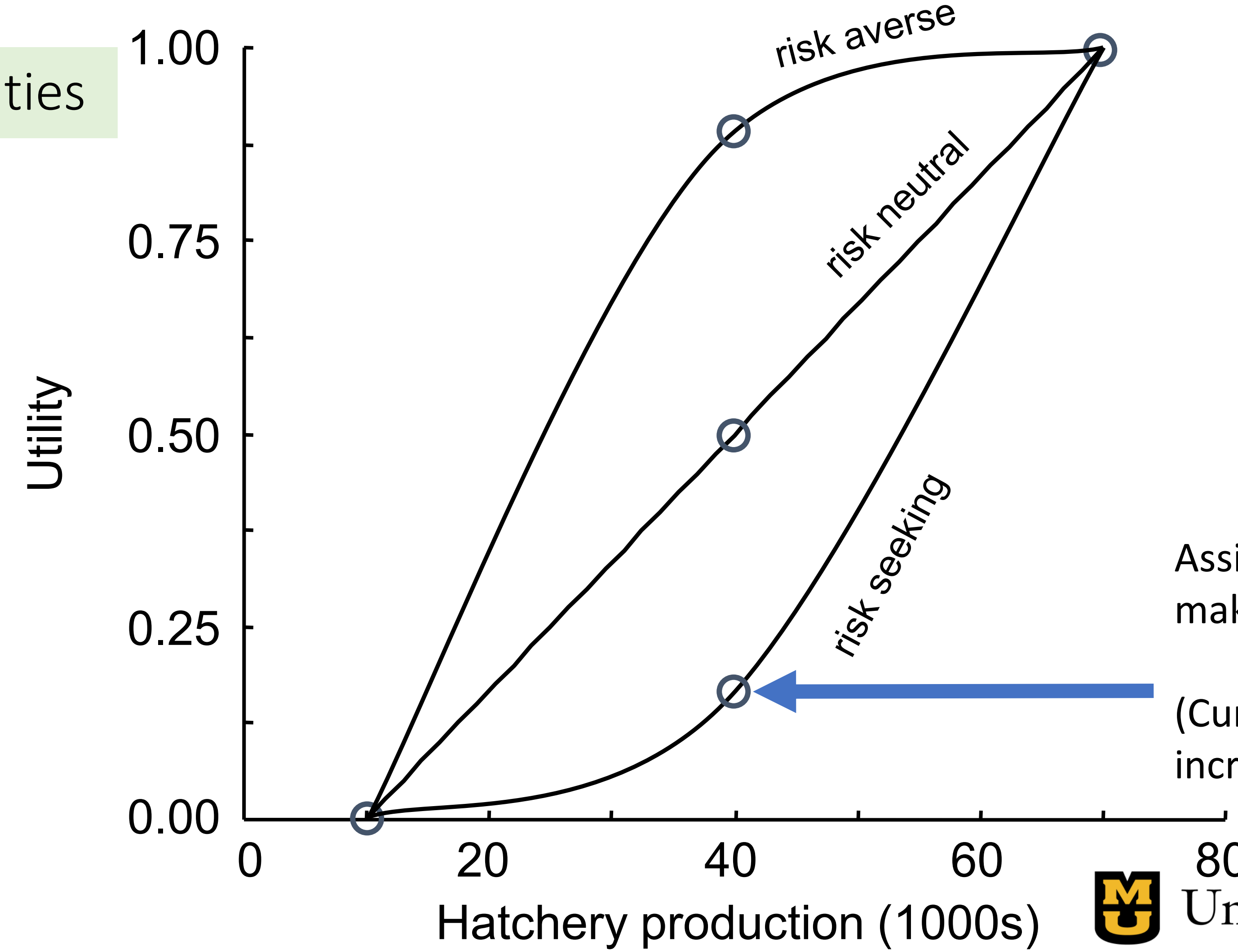
# Approaches to risk example:

- Should you adopt new technology to increase production of fry at a hatchery?

EU = assign utilities

**Utility of 40 K?????**
**40K is halfway between 10K and 70K**




Assigning EU(40K) < 0.5 makes you risk seeking

(Curve is convex showing increasing marginal utility)



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# Using Gambles to Determine Utility

- Probability equivalence:
  - What probability of getting 70K fry vs 10K fry would you view as equivalent to a sure thing of getting 40K?
  - That is, at what probability of winning would you be indifferent to the gamble versus the sure thing?
  - We can show algebraically that this probability is equal to the utility of the sure thing



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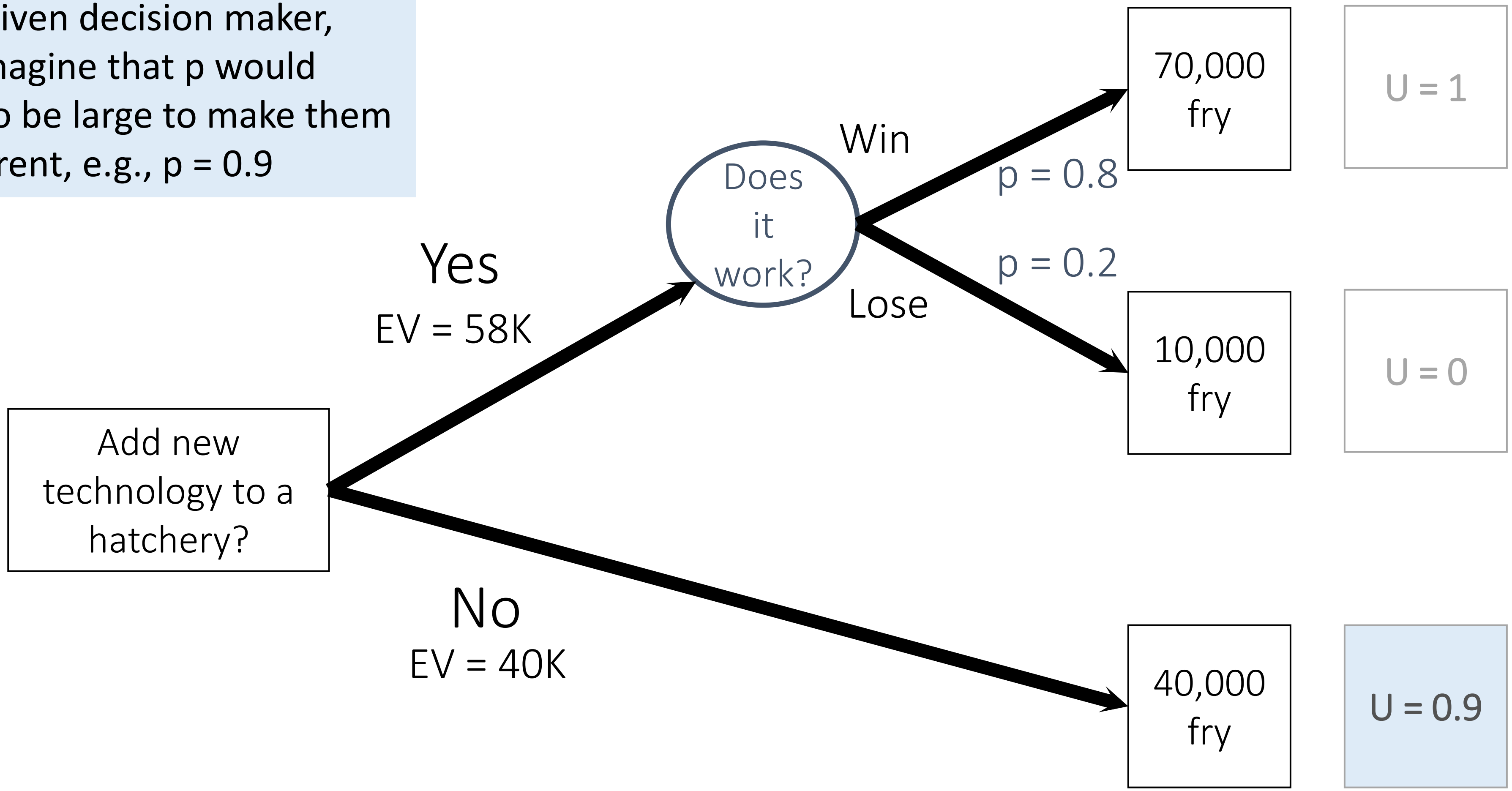
# Probability equivalence:

For a given decision maker, let's imagine that p would have to be large to make them indifferent, e.g., p = 0.9

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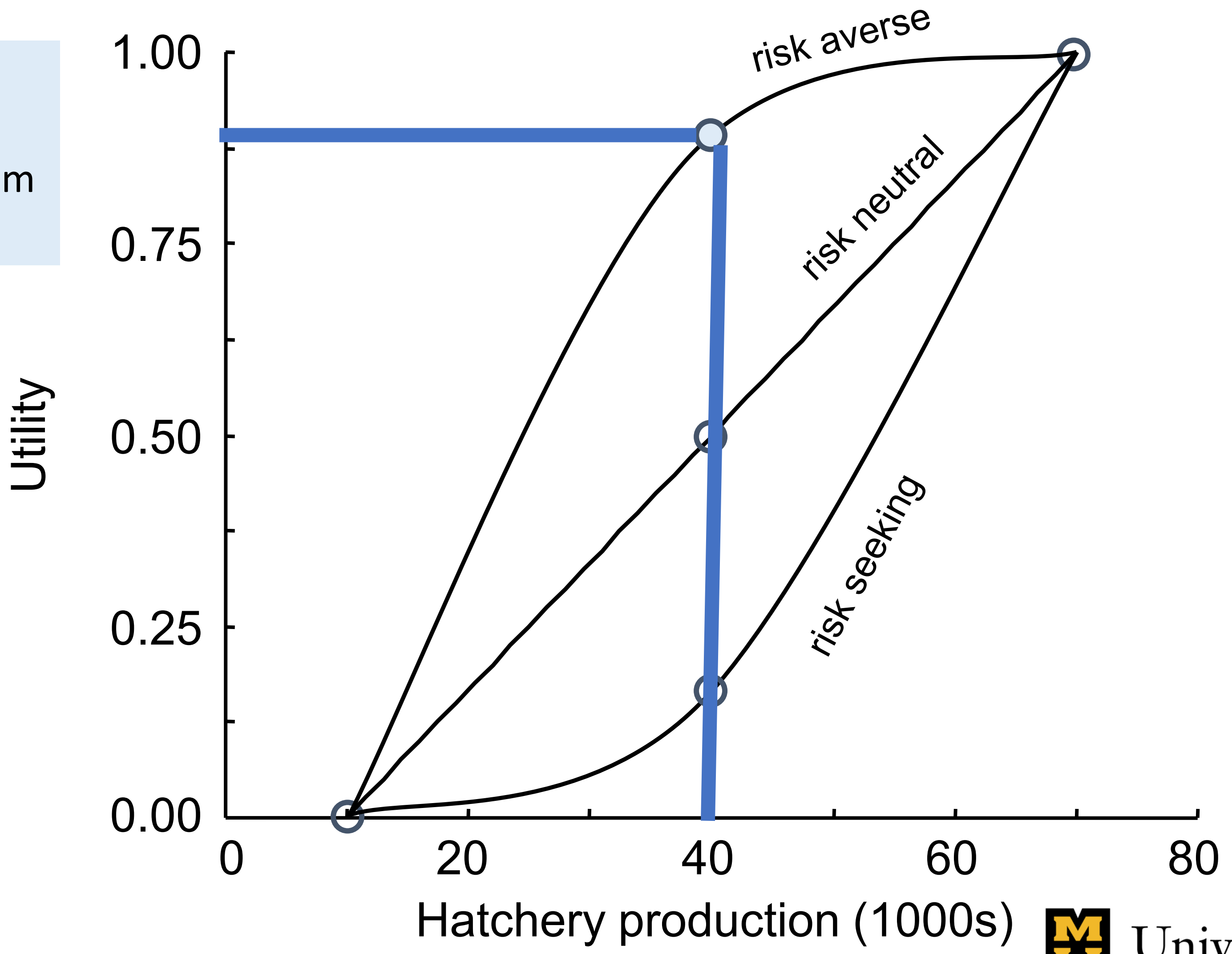
RISK AVERSE

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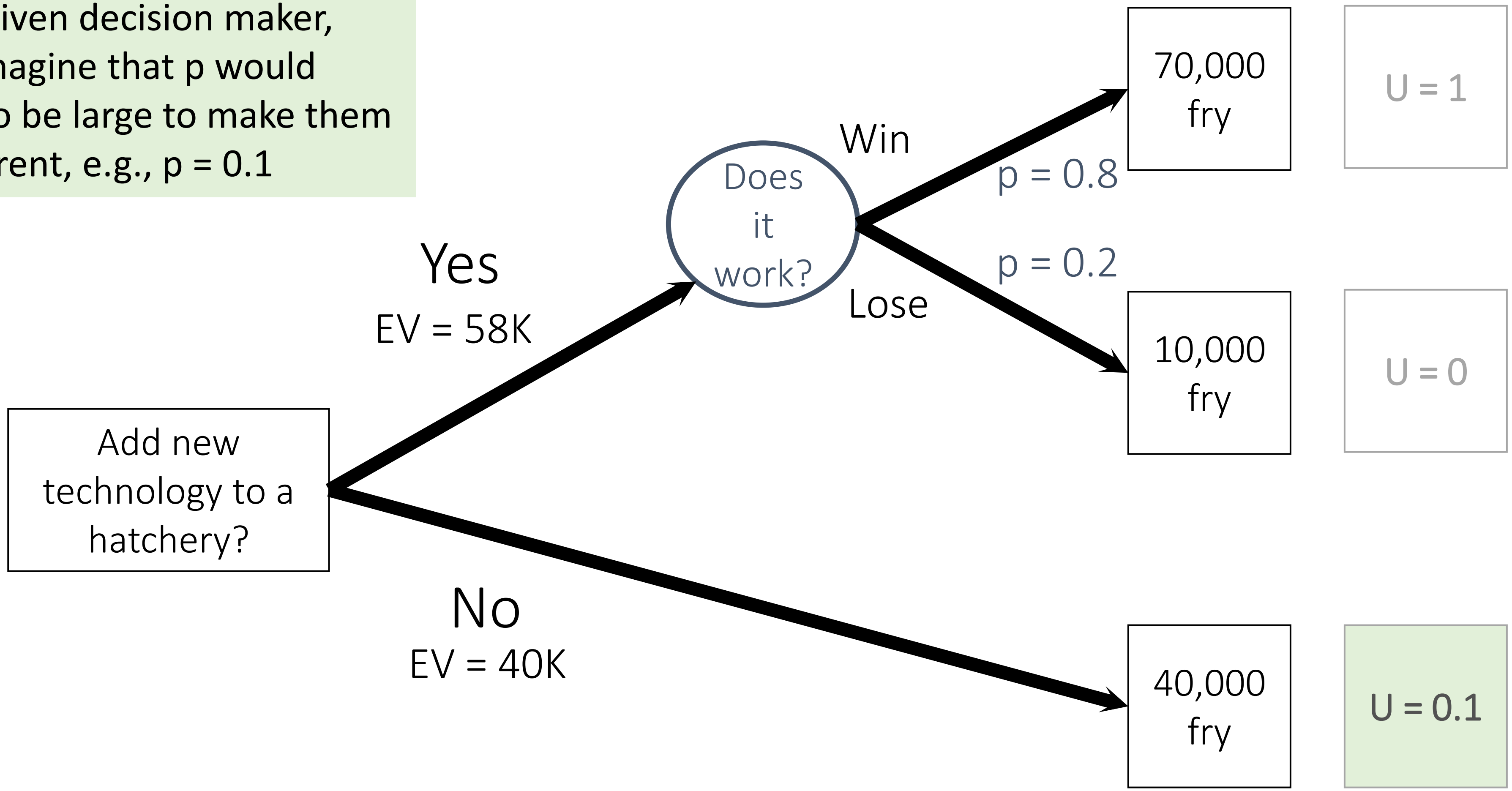
# Probability equivalence:

For a given decision maker, let's imagine that p would have to be large to make them indifferent, e.g., p = 0.1



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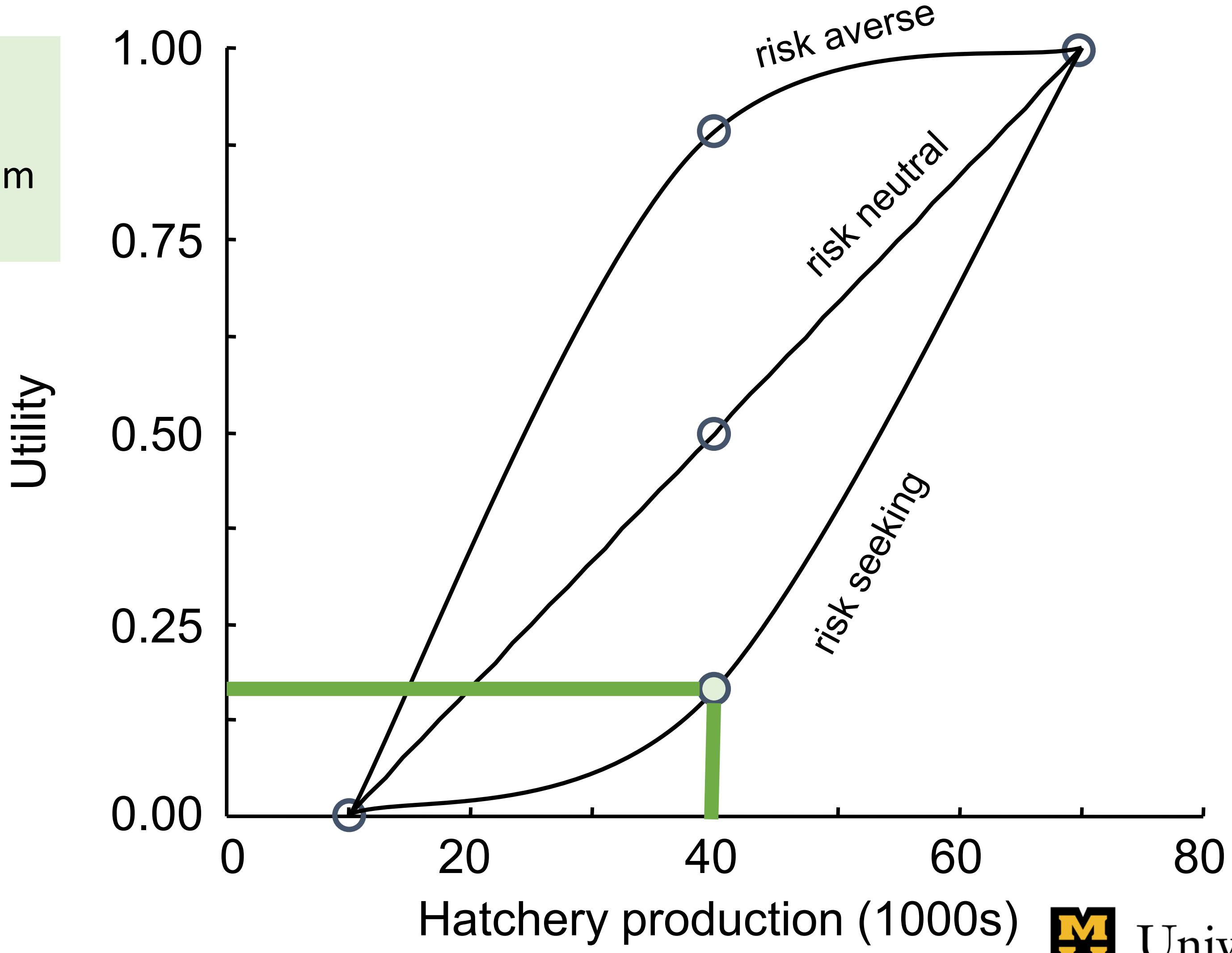
# Probability equivalence:

For a given decision maker, let's imagine that p would have to be large to make them indifferent, e.g., p = 0.1

RISK SEEKING



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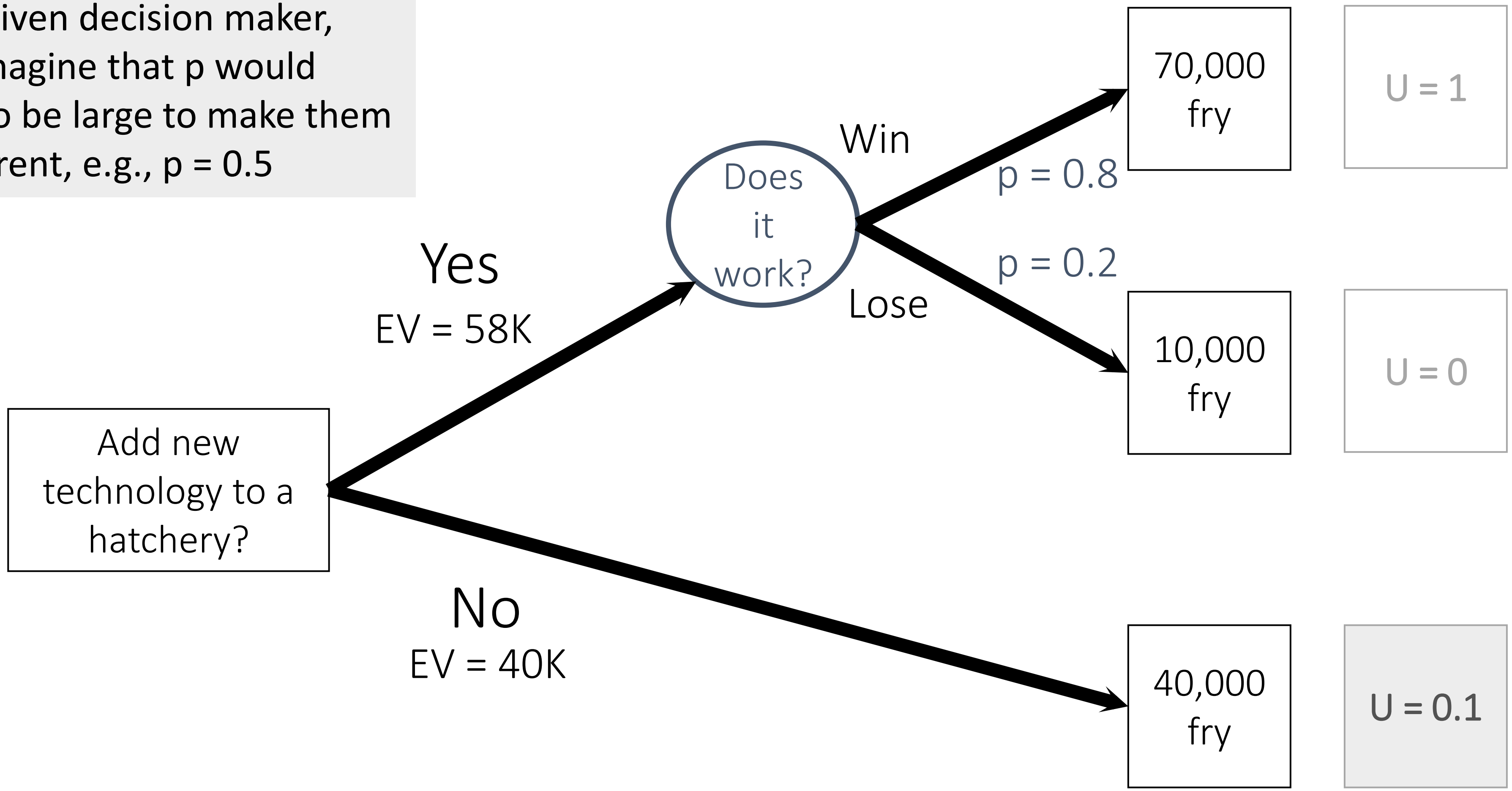
# Probability equivalence:

For a given decision maker, let's imagine that p would have to be large to make them indifferent, e.g., p = 0.5



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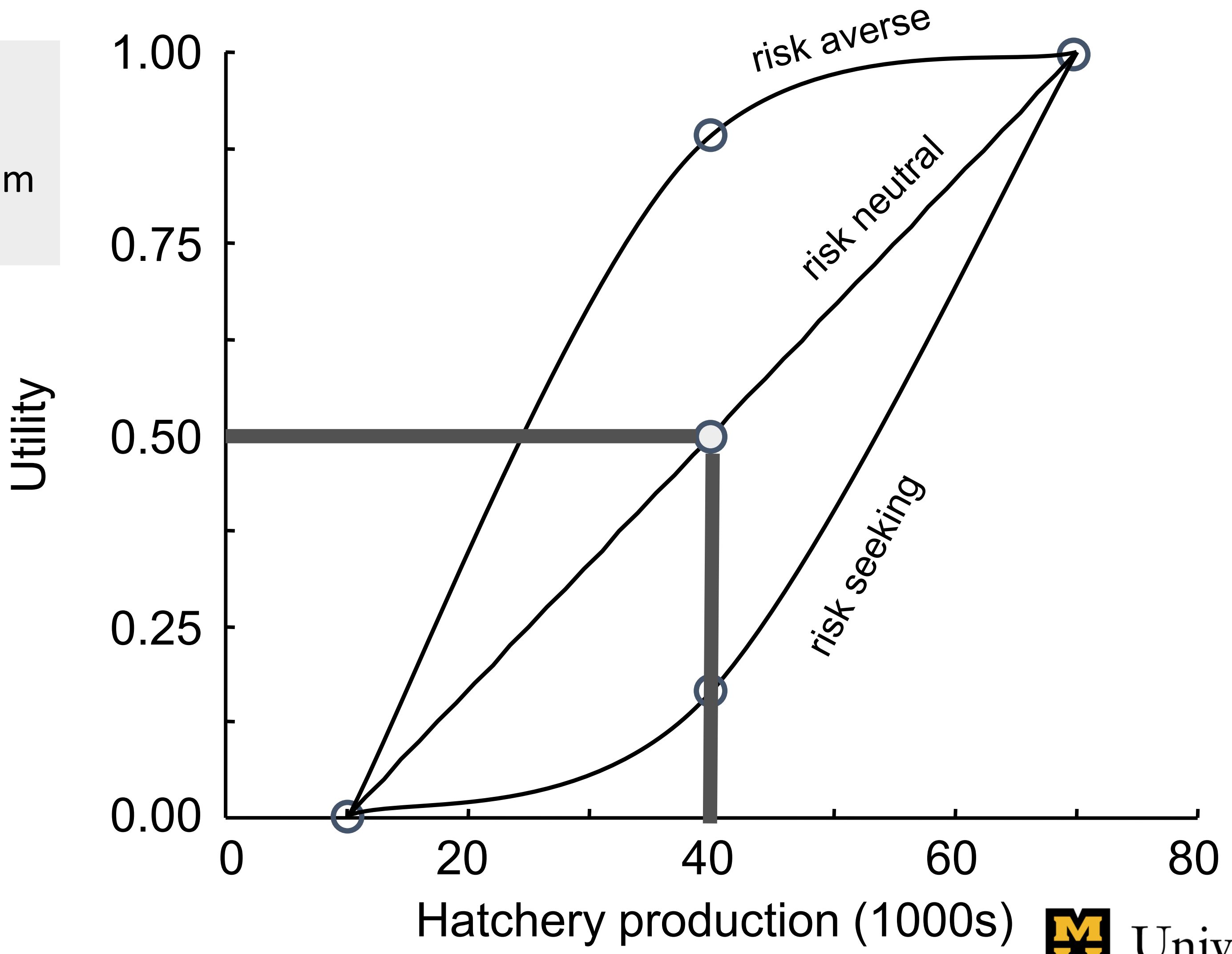
# Probability equivalence:

For a given decision maker, let's imagine that p would have to be large to make them indifferent, e.g., p = 0.5

RISK NEUTRAL = EV

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- Expected value (EV)
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  - Solution = Maximize EU
- Other approaches
  - **Mini-max:** Minimize risk of large losses
  - Robustness: Focus on attaining a minimum performance requirement over greatest range of uncertainty



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# Mini-max

- Minimize risk of large losses

**Expected value**: alternative that is best on average

*Risk-neutral decision maker*

Decision: route from St. Louis to Columbia

Expected value choice: take 70

**Mini-max**: alternative that minimizes worst possible outcome

*Risk-averse decision maker*

Decision: route from St. Louis to Columbia

Mini-max choice: avoid 70


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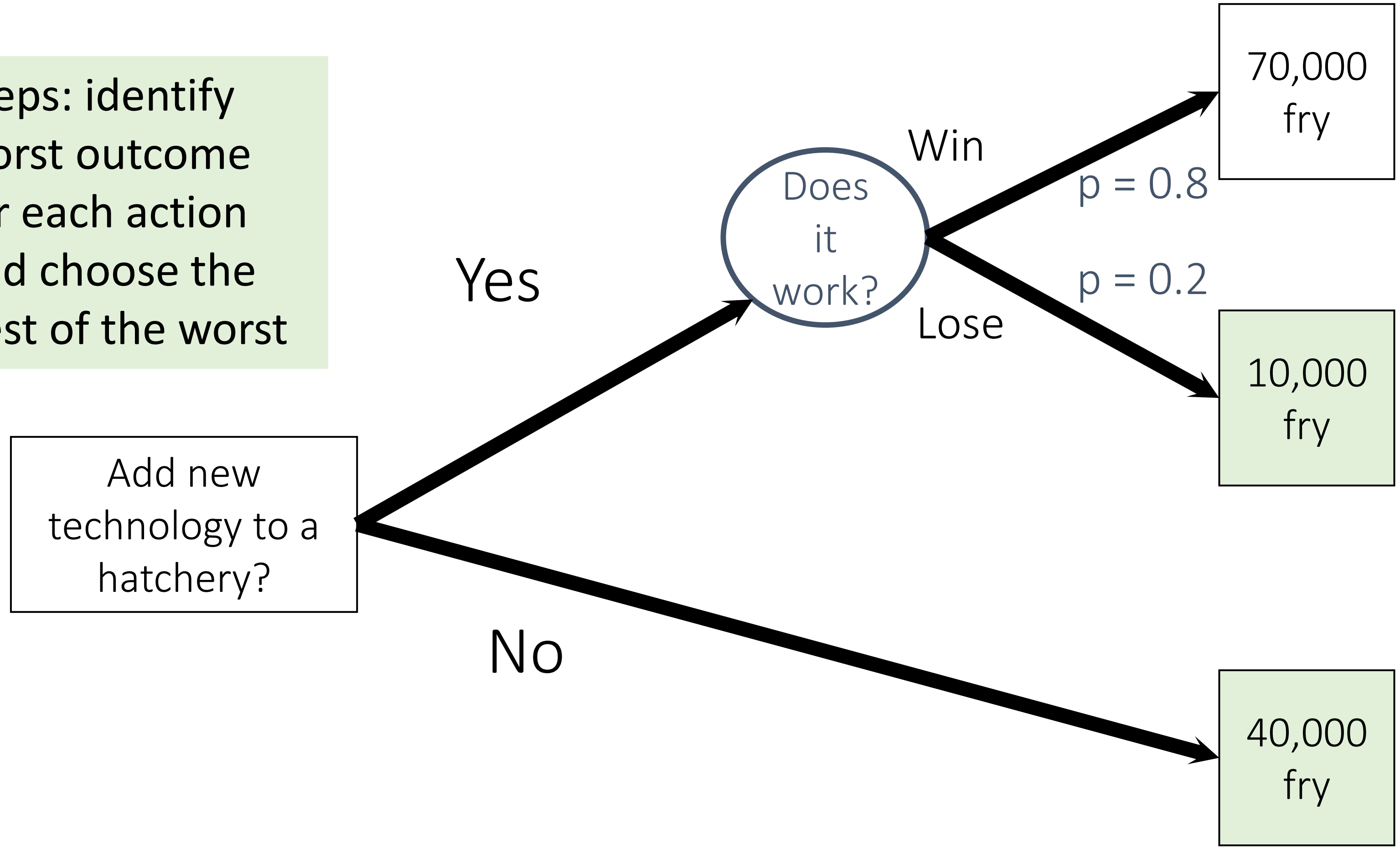
# Mini-max

- Minimize risk of large losses

Example: should you adopt new technology to increase production of fry at a hatchery?

Steps: identify worst outcome for each action and choose the best of the worst




best of the worst = 40K so we would NOT add new technology

# Mini-max vs expected value example #1

Thompson, Olden, & Converse 2024
*NeoBiota*

**Subset of the table:** management alternatives to control invasive rusty crayfish

EV

| Alternative management strategy, no. segments of removal effort | Objective (expected value) | | | Dominated by X Alternative |
|---|---|---|---|---|
| | Suppression (in millions) | Containment (%) | Prevention (in millions) | |
| Abundance, 8 | **16.67 M** | 85.7% | 1.02 M | None |
| Growth, 8 | 18.34 M | 83.1% | 0.58 M | Downstream, 8 |
| Edges, 8 | 17.92 M | 85.1% | 0.31 M | Downstream, 8 |
| Downstream, 8 | 17.32 M | **81.4%** | **0.15 M** | None |
| Random, 8 | 16.93 M | 85.7% | 0.83 M | None |

Mini-max

| Alternative management strategy, no. segments of removal effort | Objective (minimum value) | | | Dominated by X Alternative |
|---|---|---|---|---|
| | Suppression (in millions) | Containment (%) | Prevention (in millions) | |
| Abundance, 8 | 72.32 M | 100% | 4.53 M | Edges, 8 & Downstream, 8 |
| Growth, 8 | **69.91 M** | 100% | 5.00 M | None |
| Edges, 8 | 70.08 M | 100% | 2.10 M | None |
| Downstream, 8 | 71.49 M | 100% | **1.70 M** | None |
| Random, 8 | 73.30 M | 100% | 5.72 M | All |

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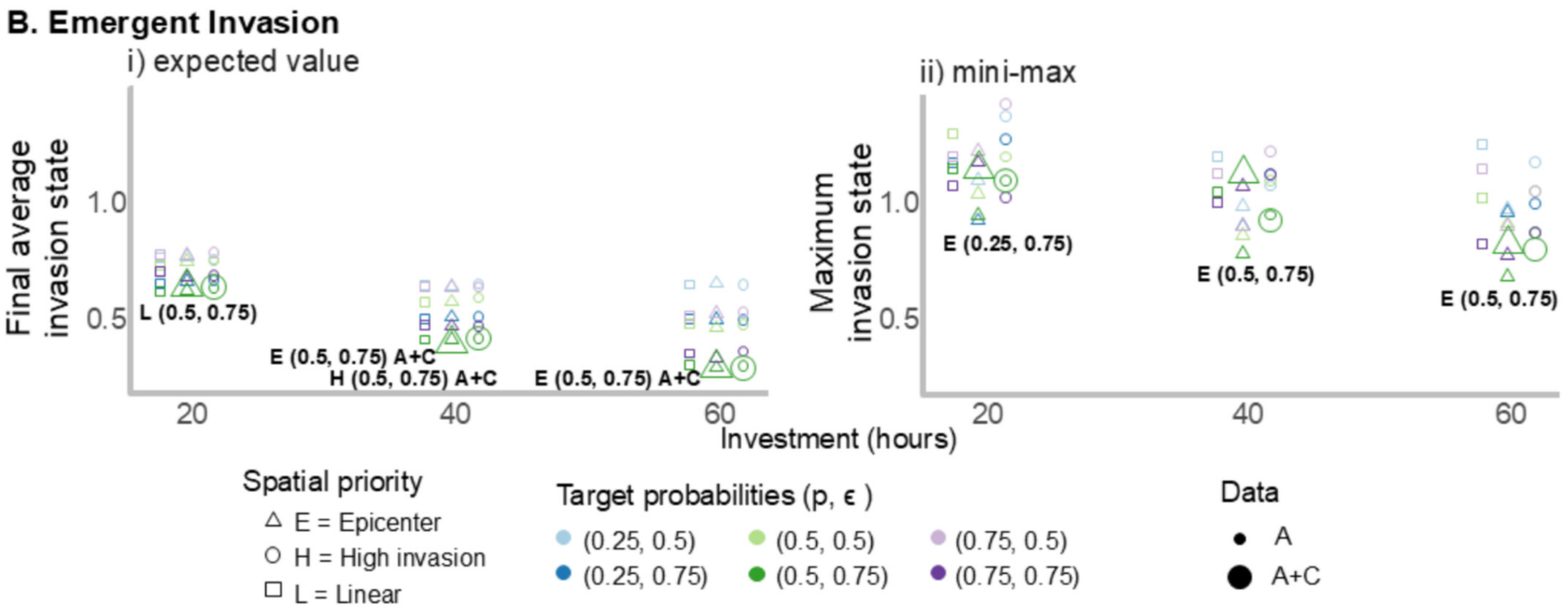
# Mini-max vs expected value example #2

**Management outcomes for invasive flowering rush**

Thompson et al. 2025

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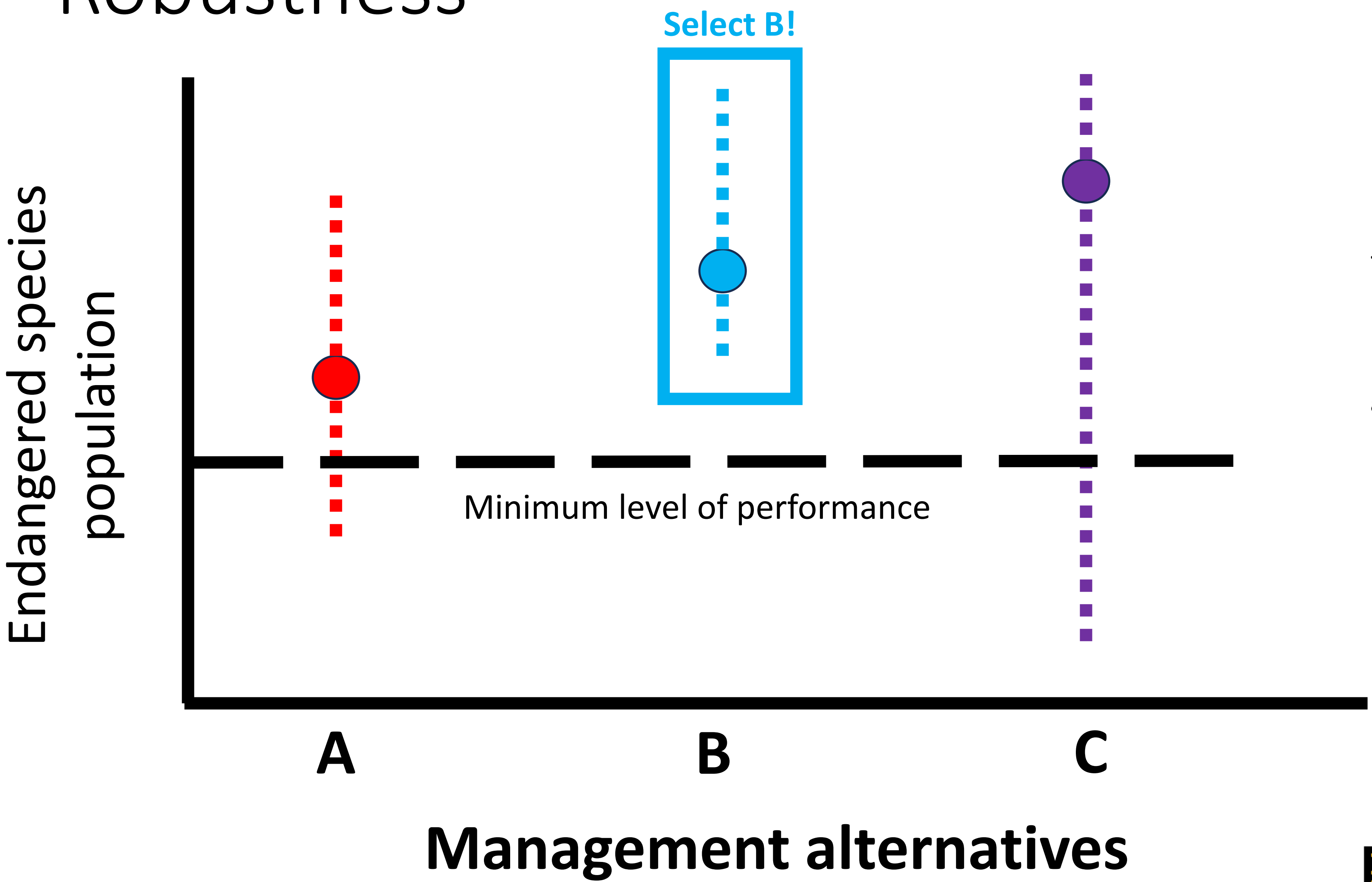
# Robustness

- Robust-satisficing: focus on attaining a minimum performance requirement over greatest range of uncertainty

  "Decision maker defines a minimum level of performance they consider to be satisfactory and chooses the option that ensures at least that level of performance over the greatest range of uncertainty"

  *Ben-Haim 2006

# Robustness




"Decision maker defines a minimum level of performance they consider to be satisfactory and chooses the option that ensures at least that level of performance over the greatest range of uncertainty" (Runge et al. 2020 Ch 13)



Developed by Brielle Thompson

# Skills Check: Risk attitudes

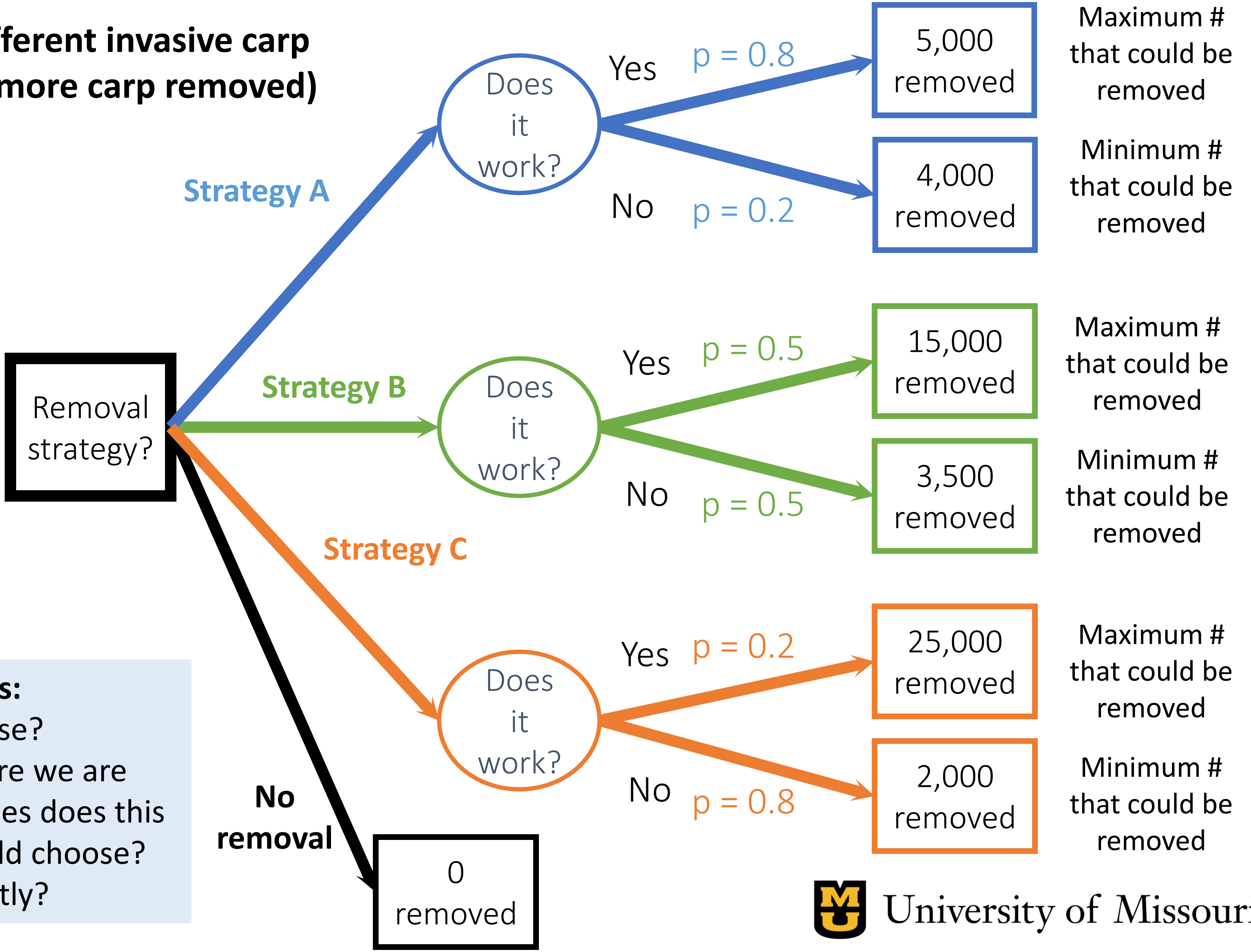
**You are choosing between different invasive carp removal strategies (you want more carp removed)**

**Across removal strategies identify:**

1) The strategy for a risk-neutral decision maker (the highest EV)
2) The strategy for a risk-averse decision maker (best mini-max)
3) Identify the strategy for a risk-seeker

**& Answer the questions:**

- Which strategy would you choose?
- What if this was a decision where we are reintroducing endangered species does this change which strategy you would choose? Would you address risk differently?

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# Tools to handle identify and deal with uncertainty:

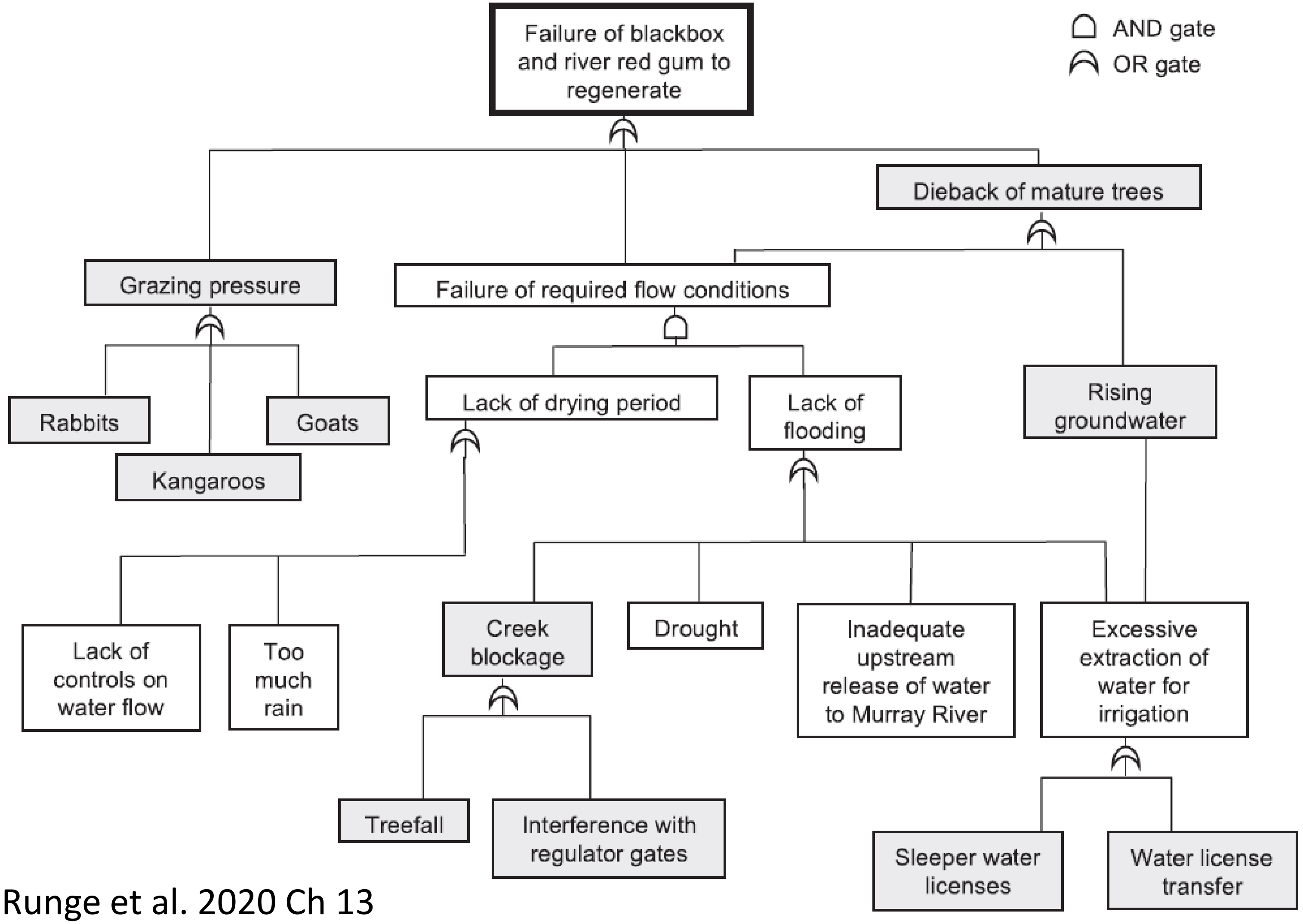
- Decision trees
- Fault trees



Developed by Brielle Thompson, see Runge et al. 2020 Ch 13

# Fault trees

- Infrequently used




Runge et al. 2020 Ch 13

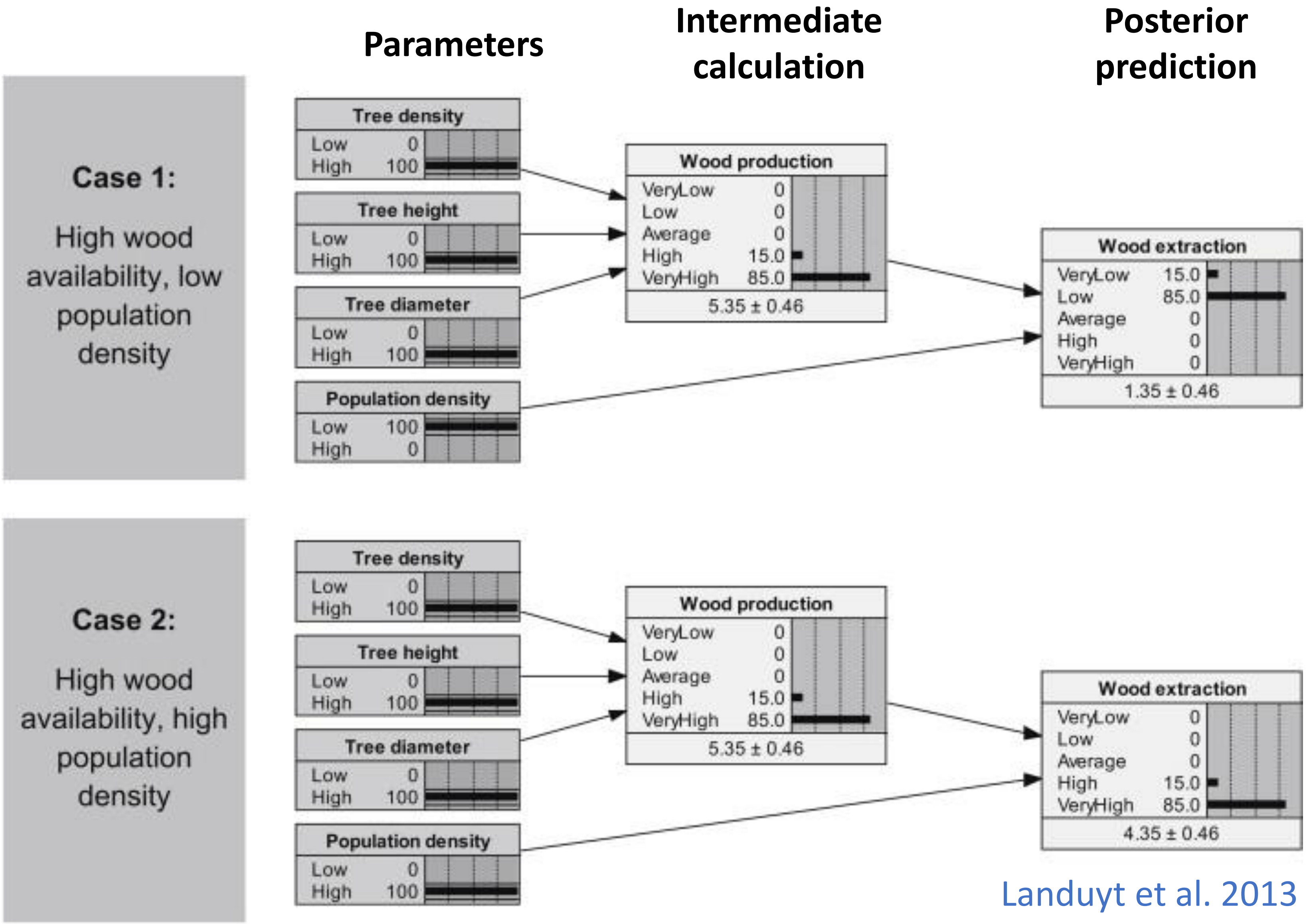
*Figure 13.1.* A fault tree illustrating various events that can lead to failure of black box and river red gum trees to regenerate. Each box represents a separate type of event, and the hierarchical structure shows how those events combine to lead to the top-level failure. "OR gates," such as that immediately below the top-level failure box, indicate that any 1 of the events below is sufficient to lead to the failure. "AND gates" indicate that all of the lower-level events must occur for the failure to occur. The shaded boxes were added in a second iteration, demonstrating the value of prototyping. *Example from Hattah-Kulkyne National Park, Victoria, Australia, Carey et al. 2005.*

# Tools to handle identify and deal with uncertainty:

- Decision trees
- Fault trees
- Bayesian belief nets

# Bayesian Belief Network (BBN) models

- Model that incorporates **Bayesian inference** to estimate uncertainty in its predictions
  - Incorporates data, prior distributions, likelihood functions




Landuyt et al. 2013



Developed by Brielle Thompson, see Landuyt et al. 2013

# Tools to handle identify and deal with uncertainty:

- Decision trees
- Fault trees
- Bayesian belief nets
- Integrated population models
  - Integrate data streams to better estimate parameter values
- Expert judgement methods
- Expected value
- Expected utility analysis (utilities elicited via gambles)
- Maxi-min criterion
- Robust satisficing
- Last week (Value of information, adaptive management, sensitivity analysis)

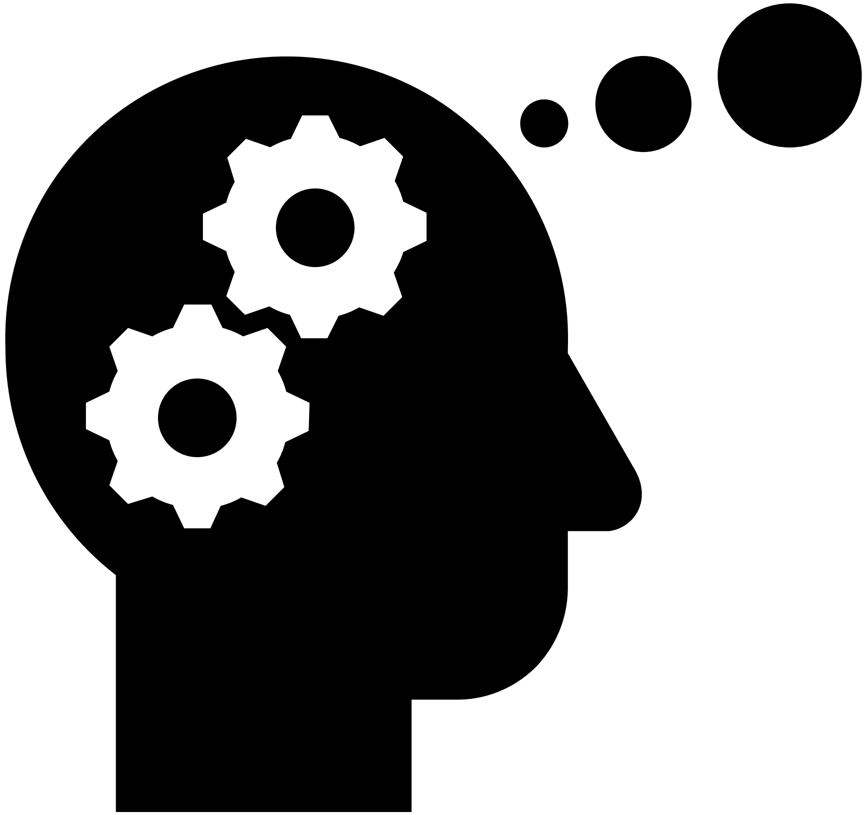


Developed by Brielle Thompson, see Runge et al. 2020 Ch 13

# Risk summary

- Decisions in the face of uncertainty
  - Need to consider decision maker’s attitude toward risk
  - This amounts to valuing outcomes in a way that is not necessarily linear relative to gain or loss
  - A utility function can be used to capture the decision maker’s risk attitude

# Discussion



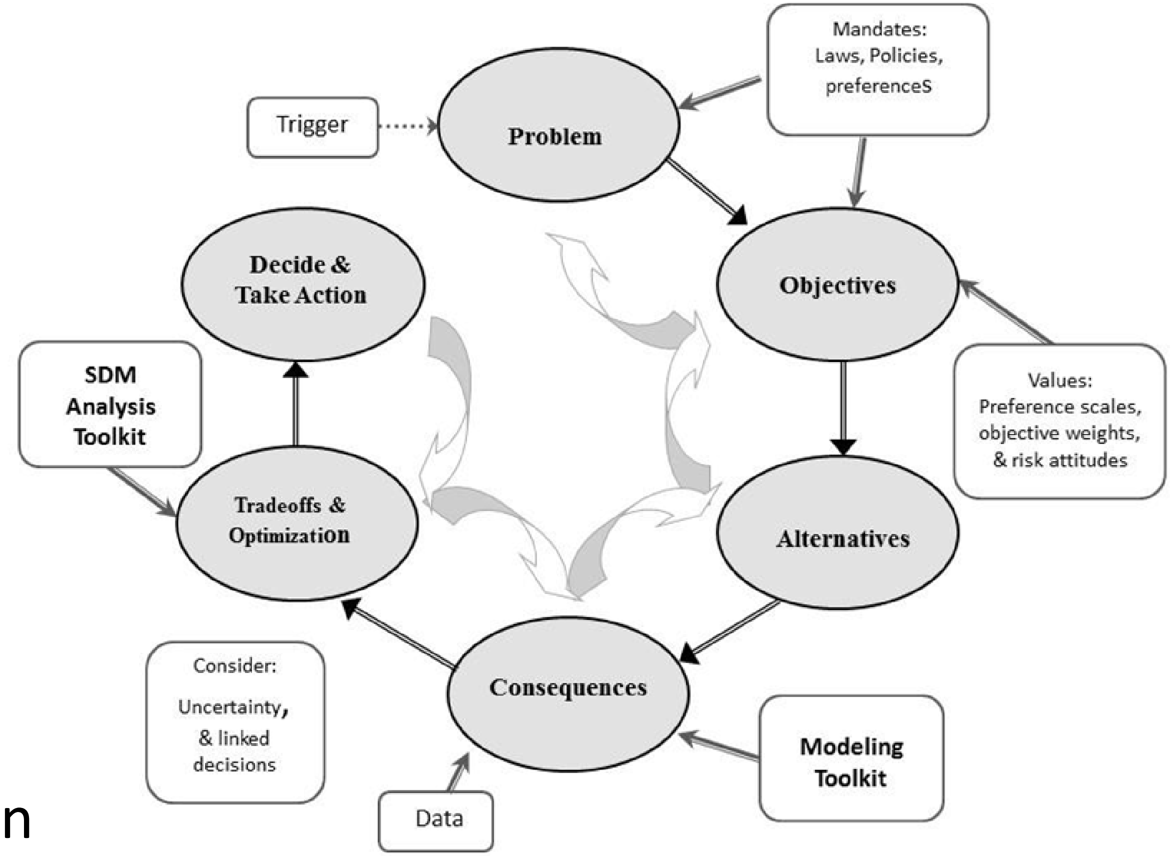
- Why should risk tolerance be understood?
- In what ways might a risk-averse decision maker approach a conservation problem differently than a risk-seeking one?

# Closing thoughts – what **did** you learn?

## Two key elements of Structured Decision Making

**1. Values-focused**
- Objectives are discussed first
- Contrasts with alternative-focused methods

**2. Problem decomposition (PrOACT)**
- Break problem into components, separating science from values
- Complete relevant analysis
- Recompose the parts to make a decision
- <u>Additional considerations</u>: risk, uncertainty, adaptive management




Source: Jean Fitts Cochrane

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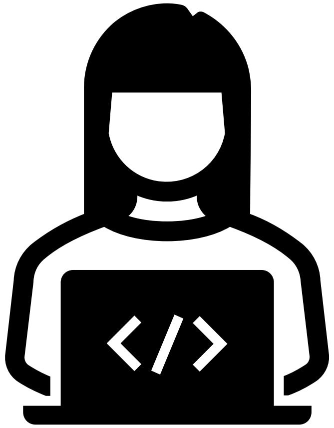
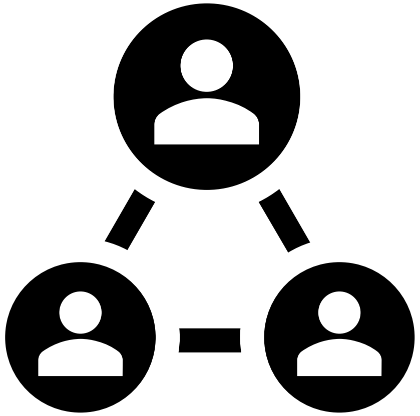
Developed by Ashley Fortune Isham, Jim Lyons, Sarah J Converse, Brielle Thompson

# Closing thoughts – what **didn’t** you learn?

- What we didn’t cover:
  - **Variety of quantitative tools**
    - Other types of value of information analysis
    - Multi-Criteria Decision analysis tools
    - Bayesian Belief Networks
    - Adaptive management modeling techniques
    - Projection modeling
    - Optimization modeling
  - **Dealing with people**
    - Stakeholder analysis, forming a team
    - Facilitation
    - Expert elicitation





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# Closing thoughts – What can SDM be used for?

Waterfowl harvests
(Williams and Johnson 1995)

Bighorn Sheep disease mitigation
(Sells et al. 2016)

Whooping crane management
(Moore et al. 2008)

Dreissenid mussel management
(Sepulveda et al. 2022)

Bull trout reintroduction
(Brignon et al. 2017)

Prairie Pothole wetland management
(Hunt et al. 2020)

## Across many scales

- Routine, one decision maker
- Small-scale, one decision maker or closely collaborating decision makers, few stakeholders
- Medium-scale, multiple collaborative decision makers, many stakeholders
- Large-scale, multiple collaborative decision makers, many stakeholders

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Developed by Brielle Thompson, Michael C. Runge

# Closing thoughts – What can SDM be used for **beyond natural resources**?

Buying a car

Choosing a college

Career decisions

Deciding dinner plans

Buying a house

Making travel plans

Developed by Brielle Thompson

University of Missouri

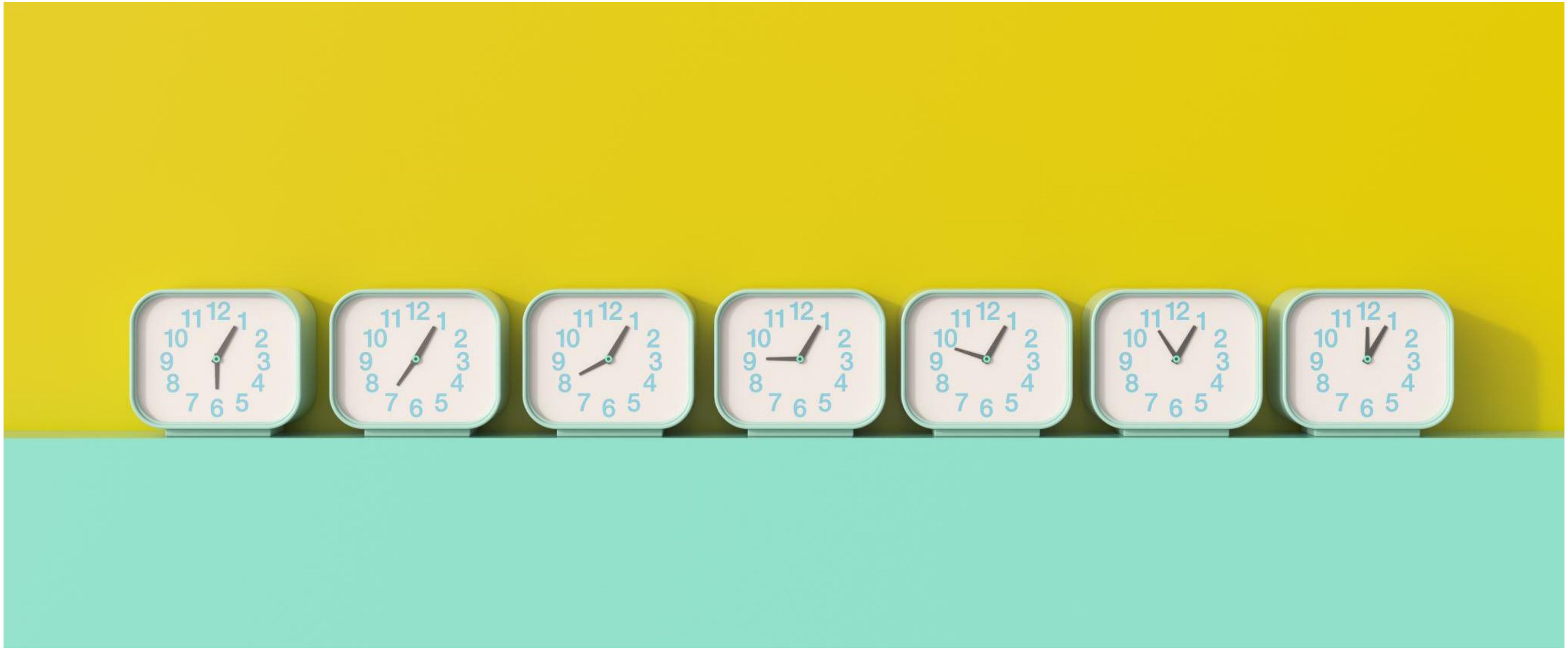
# Next week:

final class period
2-4pm ABNR 210

Final presentations!

University of Missouri

# Extra time activities:



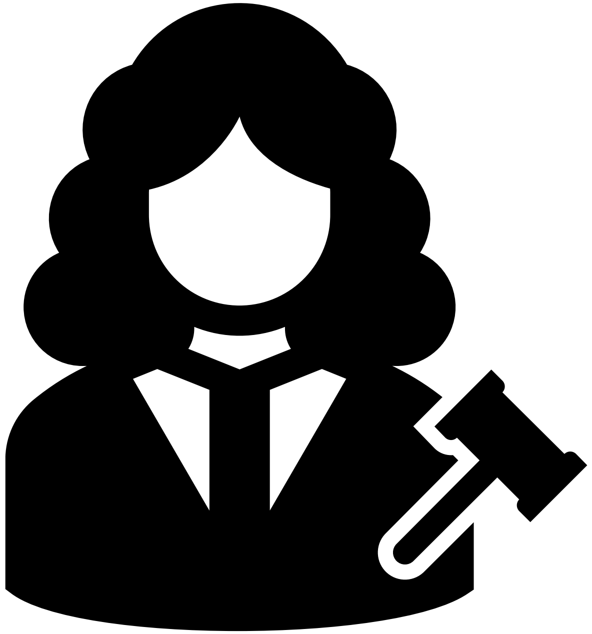
# Activity: Risk attitude debate

In the skills check exercise on invasive carp removal strategy B was the best for a risk neutral decision maker, strategy A was the best for a risk-averse decision maker, and strategy C was the best for a risk-seeker.

- Let’s have a risk attitude debate!
  - Group 1 decision maker: Risk-neutral
  - Group 2 decision maker: Risk-averse
  - Group 3 decision maker: Risk-seeking

**Instructions:**

- You have 5-10 minutes to produce your arguments
- You will have one spokesperson per team for round 1 of arguments
- A separate spokesperson will argue for round 2 -rebuttals



# Activity: Risk Discussion

- **Scenario 1**: You can either reintroduce an endangered species now with limited data or wait 3 years for better data but risk habitat degradation
  - Option A: Reintroduce now
  - Option B: Wait for data
- **Scenario 2**: You can invest in a new monitoring technology that might double detection rates, but it could fail completely. Or you can continue to use your current monitoring technology which produces worse detection but is consistent
  - Option A: Invest in new technology
  - Option B: Use current technology
- **Scenario 3**: You can implement a prescribed burn now to reduce wildfire risk, but weather conditions are marginal and could lead to unintended spread. Waiting for ideal conditions may delay the burn for months.
  - Option A: Burn now
  - Option B: Wait

For each scenario would you rather choose option A or B. Why?

# Reading discussion (Runge et al. 2020 Ch 13)

1. How does risk tolerance influence the choice between management actions with uncertain outcomes?
2. How does expected utility theory help integrate scientific uncertainty with stakeholder values? What challenges might arise when eliciting utility functions from decision makers?
3. How might cognitive biases affect how decision makers perceive and respond to risk in conservation settings? What strategies can help mitigate these biases?
4. In what ways can structured decision making tools help clarify value-based disagreements among stakeholders in conservation planning?